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A Dissertation Phase I Report on

SELF-SUPERVISED DOMAIN ADAPTATION FOR SATELLITE IMAGERY

Submitted in partial fulfillment of the requirements for the degree of

MASTER OF TECHNOLOGY

in

Computer Science

by

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Declaration

I, hereby declare that the Dissertation Phase I work titled "**Self-Supervised Domain Adaptation for Satellite Imagery**" is a record of original project work undertaken for the award of the degree of **Master of Technology** in **Computer Science**. I have completed this study under the supervision of **Prof. Benedict Tephila M**, Department of Computer Science and Engineering and **Dr. Naveen J**, Department of Computer Science and Engineering.

I also declare that this project report has not been submitted for the award of any degree, diploma, associateship, fellowship or other title anywhere else. It has not been sent for any publication or presentation purpose.

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Abstract

Deep learning models for satellite imagery degrade when they are applied to a sensor other than the one they were trained on. Existing translation-based remedies need domain labels, are limited to two domains, or handle one target at a time. This report addresses the first two objectives of a project on self-supervised domain adaptation for satellite imagery: a domain encoder that learns continuous embeddings from unlabeled multi-sensor patches, and a single conditional generator that translates a patch into the style of an arbitrary target domain.

The encoder is trained with a contrastive objective on augmented views of unlabeled patches and maps each patch to a 64-dimensional unit vector. The generator is one network that is conditioned on this vector through adaptive instance normalization and serves all ordered domain pairs; it is trained adversarially with a style-projection discriminator and with style, cycle-consistency, semantic, identity and edge losses. Both networks were trained on a CPU with 1,200 patches per sensor from PASTIS-HD (Sentinel-2, SPOT 6/7 and Sentinel-1 SAR) and evaluated on 300 held-out patches per sensor from a largely different region.

The encoder solves its pretext task (held-out loss 0.24 against 2.08 for an uninformative embedding) and produces a continuous embedding, but it identifies the sensor only moderately (65.8% against 33.3% by chance). The generator preserves scene structure (edge correlation 0.71 to 0.94, against 0.61 to 0.70 between real sensor pairs) and responds smoothly to interpolated styles for the pairs that involve SAR. It does not reproduce the radiometry of the target sensor: a constant image agrees better with the real target patch than the translation does, and translations into Sentinel-2 collapse to a source-independent color cast. The report also documents properties of the data conversion that affect such experiments, namely a region-separating split, modalities that were not acquired at the same time, and a saturated SAR channel. Generalization to unseen sensor pairs and downstream task accuracy, which are the remaining objectives, are not evaluated here.

***Keywords*:** Domain adaptation, self-supervised learning, contrastive learning, generative adversarial networks, adaptive instance normalization, multi-sensor satellite imagery, PASTIS-HD

Contents

CERTIFICATE iii

BONAFIDE CERTIFICATE iv

ACKNOWLEDGEMENT v

DECLARATION vi

ABSTRACT vii

LIST OF FIGURES xi

LIST OF TABLES xii

GLOSSARY xiv

1 INTRODUCTION 1
- 1.1 Context and Background Study . . . 1
- 1.2 Need Analysis . . . 3
- 1.3 Problem Identification . . . 4
- 1.4 Problem Formulation . . . 4

2 LITERATURE SURVEY AND OBJECTIVES 6
- 2.1 Critical Review of Literature . . . 6
  - 2.1.1 Surveys of domain adaptation for remote sensing . . . 6
  - 2.1.2 Self-supervised representation learning for satellite imagery . 7
  - 2.1.3 GAN-based image translation across domains . . . 8
  - 2.1.4 Label-free and multi-domain translation for satellite imagery 8
  - 2.1.5 Datasets . . . 9
- 2.2 Literature Collection & Segregation . . . 10
- 2.3 Summary of Literature . . . 10
- 2.4 Identification of Gaps . . . 11
- 2.5 Need for Research . . . 12
- 2.6 Problem Statement . . . 13
- 2.7 Objectives . . . 13

2.8 Limitations of the Study 14

3 PROPOSED METHODOLOGY 15
3.1 Problem formulation 16
3.2 Proposed research framework 16
3.3 Data collection and preprocessing 17
3.3.1 Datasets in the project plan and datasets used 17
3.3.2 PASTIS-HD 18
3.3.3 Conversion to a unified multi-sensor format 19
3.3.4 Subset, split and temporal alignment 19
3.3.5 Augmentation and synthetic data 20
3.4 Algorithm & Model design 20
3.4.1 Domain encoder 20
3.4.2 Contrastive pretext task 21
3.4.3 Unified conditional generator 22
3.4.4 Discriminator with style projection 23
3.4.5 Composite training objective 23
3.4.6 Training procedure and hyperparameters 24
3.4.7 Evaluation protocol 26
3.5 System requirements 27

4 RESULTS AND DISCUSSIONS 28
4.1 Major findings from Literature 28
4.2 Summary of the proposed design 29
4.3 Implementation and experimental setup 29
4.4 Characteristics of the converted data 29
4.5 Results for Objective 1: the domain encoder 31
4.6 Results for Objective 2: the conditional generator 33
4.6.1 Qualitative results 34
4.6.2 Quantitative fidelity with respect to the co-registered reference 34
4.6.3 Style transfer and content preservation 36
4.6.4 Reconstruction with the patch's own style 37
4.6.5 Brightness bias of the generator 38
4.6.6 Continuity of the style space 38
4.7 Discussion 41
4.7.1 Objective 1 41
4.7.2 Objective 2 41
4.7.3 Relation to the related work 42
4.7.4 Summary of the achievement of the objectives 42
4.8 Limitations and threats to validity 42

5 CONCLUSION 45
5.1 Summary of the work 45
5.2 Conclusions with respect to the objectives 46
5.2.1 Objective 1: domain encoder 46

5.2.2 Objective 2: conditional generator 46
5.3 Contributions 47
5.4 Scope for future research and development 47

BIBLIOGRAPHY **48**

A Layer-wise Network Specification **53**

B Code Organization and Reproducibility **55**
B.1 Organization of the code base 55
B.2 Commands 56
B.3 What can and cannot be verified from the stored artifacts 56

LIST OF FIGURES

3.1 Workflow of the research methodology. The dashed box is outside the scope of this report. 15
3.2 Two-stage framework. In Stage 1 the encoder is trained with a contrastive objective on two augmented views of unlabeled patches. In Stage 2 the encoder is frozen; it supplies the style embedding z_{tgt} of a target exemplar to the generator, which combines it with the content of a source patch. 17

4.1 Held-out embeddings projected onto their first two principal components. Small markers are patches; large outlined markers are the domain means of the projected embeddings. 33
4.2 Translations of one held-out patch per source domain (first column of each row) into the two other domains, next to the real co-registered patch of each target domain. The style is the centroid embedding of the target domain. Sentinel-1 patches are displayed as VV, VH and the saturated difference channel in red, green and blue. 35
4.3 One SPOT 6/7 patch translated with styles that are interpolated between two domain centroids (rows), for α from 0 to 1. 39
4.4 (a) Fraction of interpolated outputs that the encoder assigns to the end-point domain b; (b) mean absolute difference to the output for $\alpha = 0$. 40

LIST OF TABLES

1.1 Existing conditions and desired state for multi-sensor domain adaptation. . . . 3

2.1 Segregation of the collected literature. . . . 10
2.2 Comparison of adaptation and translation methods on the properties targeted in this project. . . . 11
2.3 Mapping of the objectives to the sections of this report. . . . 13

3.1 Framework components and the objectives they serve. . . . 18
3.2 Datasets in the project plan and their status in this report. . . . 18
3.3 Hyperparameters of the two training stages. . . . 25
3.4 Hardware and software environment. . . . 27

4.1 Main design decisions and their rationale. . . . 30
4.2 Statistics of the held-out patches after preprocessing (300 patches per domain, values in $[-1, 1]$). . . . 31
4.3 Evaluation of the domain encoder on held-out patches. . . . 32
4.4 Nearest-centroid classification of held-out patches (rows: true domain, columns: assigned domain) and cosine similarity between the domain centroids. . . . 33
4.5 Model size and inference time on the CPU (6 threads, 128×128 patches). . . . 34
4.6 Agreement with the co-registered real target patch, as PSNR in dB / SSIM (higher is better; mean over 300 held-out patches). S2 denotes Sentinel-2. . . . 35
4.7 Style transfer, content preservation and cycle reconstruction on held-out patches. "Real patches assigned" is the share of real target patches that the encoder assigns to their own domain; the value in parentheses is the edge correlation between the real source and the real target patch. S2 denotes Sentinel-2. . . . 37
4.8 Reconstruction of a held-out patch when the generator is conditioned on its own domain (PSNR in dB / SSIM). . . . 38
4.9 Mean value of each channel of the translated patches (300 patches per source domain) and of the real target patches, in $[-1, 1]$. Channels are red, green, blue and near-infrared (or proxy) for the optical domains, and VV, VH, difference and mean for SAR. . . . 39

4.10 Response of the output to the interpolation coefficient α (300 source patches). Endpoint L1 is the mean absolute difference between the outputs for $\alpha = 0$ and $\alpha = 1$; the linearity ratio is the endpoint L1 divided by the summed L1 of the four successive steps; the monotone share is the fraction of patches for which the distance to the $\alpha = 0$ output never decreases. 40
4.11 Achievement of the objectives. 43

A.1 Domain encoder E (888,224 trainable parameters). 53
A.2 Generator G (1,594,116 trainable parameters). Each AdaIN layer has two linear maps, $64 \to C$, for γ and β. 54
A.3 Discriminator D with style projection (709,601 trainable parameters). 54

GLOSSARY

Item	Description
AdaIN	**Ada**ptive **I**nstance **N**ormalization: normalization of each feature channel followed by a scale and shift computed from a style vector
DA	**D**omain **A**daptation
GAN	**G**enerative **A**dversarial **N**etwork
InfoNCE	Noise-contrastive estimation loss that maximizes agreement between positive pairs relative to negatives
kNN	**k**-**N**earest **N**eighbors
LSGAN	**L**east-**S**quares **GAN**: adversarial loss based on squared error
MLP	**M**ulti**l**ayer **P**erceptron
NIR	**N**ear-**I**nfra**r**ed band
PASTIS	**P**anoptic **A**gricultural **S**atellite **TI**me **S**eries benchmark; PASTIS-HD adds SPOT very-high-resolution imagery
PCA	**P**rincipal **C**omponent **A**nalysis
PSNR	**P**eak **S**ignal-to-**N**oise **R**atio, in decibels
SAR	**S**ynthetic **A**perture **R**adar
SE	**S**queeze-and-**E**xcitation channel attention
SSIM	**S**tructural **Sim**ilarity **I**ndex
SSL	**S**elf-**S**upervised **L**earning
UDA	**U**nsupervised **D**omain **A**daptation
VHR	**V**ery **H**igh **R**esolution
VV, VH	Radar polarizations: vertical transmit and vertical receive; vertical transmit and horizontal receive
E, G, D	Domain encoder, generator and discriminator
z, $\bar{z}_k$	Style embedding of a patch; centroid embedding of domain k
d	Embedding dimension ($d = 64$)
τ	Temperature of the contrastive loss
λ_*	Weight of a loss term
α	Interpolation coefficient between two style embeddings

Chapter 1

INTRODUCTION

This report presents the work carried out in Dissertation Phase I on self-supervised domain adaptation for satellite imagery. The overall project has four objectives; this report covers the first two, namely the design of a domain encoder that learns continuous domain embeddings without domain labels, and the development of a single conditional generator that uses these embeddings to translate imagery into the style of an arbitrary target domain. The remaining objectives, which concern generalization to unseen sensor combinations and downstream task evaluation, are outside the scope of this report and are listed only as future work.

1.1 Context and Background Study

Earth observation has moved from single-mission analysis to a regime in which many optical and radar satellites image the same regions repeatedly. Sentinel-2 provides multispectral optical imagery at resolutions between 10 and 60 m [1], Sentinel-1 provides C-band synthetic aperture radar (SAR) imagery that does not depend on illumination or cloud cover [2], Landsat-8 provides multispectral imagery at 30 m [3], and very-high-resolution (VHR) optical platforms such as SPOT 6/7 provide imagery at a native resolution of about 1.5 m [4]. Deep learning models for land-cover classification, crop-type mapping and change detection are increasingly expected to work across this heterogeneous archive rather than only on the sensor they were trained on.

A model trained on one sensor learns the statistics of that sensor. When it is applied to another sensor, the appearance of the input changes although the land-cover classes do not. This situation is known as domain shift, and learning-theoretic results show that the error on the target domain is bounded by the source error plus a term that grows with the divergence between the source and target input distributions [5]. In satellite imagery the shift has several physical causes: different spectral response functions and radiometric calibration, different spatial resolution, atmospheric and illumination conditions, and, most strongly, different imaging physics, as between an optical sensor that records reflectance and a SAR sensor that records backscatter corrupted by speckle [6, 7]. Seasonal and geographic variation add further shift even within a single sensor.

The remote sensing community has responded with a large body of domain adaptation work. Early methods selected invariant features or aligned feature spaces, and were summarized in the survey by Tuia et al. [6]. Adversarial feature alignment [8] and pixel-level image translation with cycle consistency [9, 10] later became the dominant approaches, and recent surveys organize unsupervised domain adaptation for remote sensing into generative, adversarial, self-training and hybrid families [7] and discuss the growing role of self-supervised learning [11]. In parallel, self-supervised representation learning has been applied to satellite imagery, using seasonal or geographic metadata to define positive pairs [12, 13], masked autoencoding [14], or contrastive multiview coding [15]. In the image translation literature, StarGAN [16] and StarGAN v2 [17] translate between many domains with a single generator, and for satellite imagery the label-free pairwise method SS(DA)^2 [18], the multi-source standardization method StandardGAN [19] and diffusion-based translators [20, 21] have been proposed. Chapter 2 reviews these works in detail.

The present project builds on this background. It starts from SS(DA)^2 [18], which removes the need for labels and for an explicit domain boundary but supports only two satellites at a time, and asks how the same label-free idea can be extended to many sensors within a single model. The approach taken is to separate the problem into a representation part, a domain encoder that summarizes the style of any patch as a continuous vector, and a generation part, a single generator that renders the content of a source patch in the style given by such a vector.

1.2 Need Analysis

The need for the work follows from the gap between how domain adaptation is currently practiced for satellite imagery and how it would have to work for a growing, multi-sensor archive. Table 1.1 contrasts the existing condition with the desired state along the aspects that matter for this project.

Table 1.1: Existing conditions and desired state for multi-sensor domain adaptation.

Aspect	**Existing condition**	**Desired state**
Supervision	Labeled target data, or explicit domain identities for every training image, are usually required.	Learning from unlabeled patches, with no domain labels.
Domain representation	Discrete domain indices, or an implicit pair of domains.	A continuous embedding in which any patch can be placed.
Scalability	One model per sensor pair, so the number of models grows with the number of sensor pairs.	One model that serves all sensors.
New styles	Adding a sensor means retraining.	A new sensor is embedded by the same encoder.
Content fidelity	Translation may alter or invent ground structures.	Explicit constraints that keep geometry and edges unchanged.
Computation	Some recent translators rely on iterative diffusion sampling.	A single feed-forward pass that is feasible on modest hardware.

Three practical needs stand out. First, labeled data are expensive and sensor-specific, so a method that needs none in the target domain is more useful in practice. Second, the number of sensors in use keeps growing; if a separate translator is required for every pair, the number of models grows quadratically with the number of sensors, which is not sustainable. Third, in blended multi-sensor mosaics a clear domain boundary often does not exist, so methods that require every image to carry a domain label cannot be applied directly. A continuous, label-free representation of domain addresses all three needs at once.

1.3 Problem Identification

A systematic reading of the literature and of the requirements above leads to the following specific problems, which the two objectives of this report address.

P1. Label dependence. Most translation-based adaptation methods need domain-level supervision: every training image must be assigned to a known domain, and the number of domains is fixed beforehand [16, 17, 19].

P2. Pairwise formulation. The label-free method that is closest to this work translates between exactly two satellite domains [18]. Covering K sensors then needs a separate model for each of the $K(K-1)/2$ sensor pairs.

P3. Discrete domain conditioning. Conditioning on a one-hot domain label [16] cannot express a sensor that was not enumerated at training time, and it offers no notion of similarity between sensors.

P4. Single-sensor representations. Self-supervised pretraining for remote sensing is mostly done within one sensor or sensor family [12–15], so it does not provide a representation that places different sensors in a common continuous space, and it has no generative component.

P5. Content preservation. Adversarial translation can change the geometry of a scene. For satellite imagery, where field boundaries, roads and water bodies carry the information, translation must be explicitly constrained to keep them.

1.4 Problem Formulation

The choice of the project topic followed from the base paper. SS(DA)^2 [18] shows that domain adaptation for satellite images can be done without labels and without a predefined domain boundary, which removes problem P1. Its limitation to two domains (P2) suggested the direction of the project: keep the label-free training signal, but make the domain representation continuous and let a single generator use it. StarGAN v2 [17] provided the architectural template for such a generator, because it injects a style code into the generator through adaptive instance normalization, but it obtains its style codes from labeled domains. The

thought process therefore was to replace the labeled style code by an embedding that is learned by a self-supervised encoder from unlabeled multi-sensor patches.

This decomposition naturally gave four objectives, of which the first two are the prerequisites for the others: a domain encoder (Objective 1) and a conditional generator (Objective 2) have to exist before generalization to unseen sensor pairs and downstream accuracy can be studied. The project plan presented in the earlier reviews also included a pairwise baseline, so that a working result would exist even if the multi-domain model did not fully converge; it is retained as a comparison for the later objectives. The formal problem statement is given in Section 2.6, and its mathematical form in Section 3.1.

The report is organized as follows. Chapter 2 reviews the literature, identifies the research gap and states the objectives. Chapter 3 describes the proposed framework, the data and the models. Chapter 4 summarizes the findings from the literature, describes the implementation and reports and discusses the results obtained for the two objectives. Chapter 5 concludes and lists the scope for future work.

Chapter 2

LITERATURE SURVEY AND OBJECTIVES

This chapter reviews the literature that is relevant to a label-free, multi-domain adaptation framework for satellite imagery. The works were collected during the initial project reviews and grouped into five categories: surveys of domain adaptation, self-supervised representation learning for remote sensing, GAN-based image translation, label-free and multi-domain translation methods for satellite imagery, and datasets. The chapter ends with the identified gaps, the problem statement and the two objectives that this report addresses.

2.1 Critical Review of Literature

2.1.1 Surveys of domain adaptation for remote sensing

Tuia et al. [6] give one of the first structured overviews of domain adaptation for the classification of remote sensing data. They organize the methods that existed at the time into the selection of invariant features, the adaptation of feature-space distributions, the adaptation of classifiers, and active learning that selects new labeled samples in the target domain. Their scope is image classification, and the survey predates generative adversarial and self-supervised methods, so it does not cover contrastive embeddings or generative cross-sensor translation. Xu et al. [7] review unsupervised domain adaptation methods that are based on remote sensing

data and classify them into generative, adversarial, self-training and hybrid approaches. They also define the partial, open-set and multi-domain settings. The multi-domain setting is described conceptually, but the survey proposes no method for it and reports no empirical test of generalization to unseen sensor pairs. Khazri Khelif et al. [11] focus on domain adaptation for satellite images that is based on self-supervised learning, and discuss advances, challenges and perspectives. Like the other two, it is a survey, so it does not propose an architecture or evaluate generalization to unseen sensor combinations. These surveys establish the taxonomy and the open problems, and they motivate the multi-domain, self-supervised direction of this project.

2.1.2 Self-supervised representation learning for satellite imagery

Contrastive learning methods such as SimCLR [22] and MoCo [23] learn representations by pulling together the embeddings of two augmented views of the same instance and pushing apart the embeddings of other instances, using an objective of the InfoNCE type [24]. Several works have adapted this idea to satellite imagery. Seasonal Contrast (SeCo) [12] follows the MoCo scheme and treats images of the same location taken at different times as positive pairs, pretraining on uncurated global Sentinel-2 data. Geography-aware self-supervised learning (GASSL) [13] uses geolocation and temporal metadata to define positive pairs and adds a geolocation prediction task. SatMAE [14] is a masked autoencoder with a vision transformer backbone and temporal and spectral positional encodings, which learns by reconstructing masked patches. Stojnić and Risojević [15] apply contrastive multiview coding to remote sensing scene classification and study how the choice of pretraining data affects downstream accuracy. All four works show that pretraining on remote sensing imagery is beneficial, but they operate within one sensor or sensor family and are evaluated on classification tasks. They do not treat the difference between sensors, for example between optical and SAR imagery, as the quantity to be represented, they do not produce a continuous embedding space of sensors, and they have no generative component. The encoder of this project borrows the contrastive objective from this line of work but uses it to represent the style of a patch across several sensors.

2.1.3 GAN-based image translation across domains

Generative adversarial networks [25] are the basis of most learned image translation. CycleGAN [9] translates between two domains without paired examples by combining an adversarial loss with a cycle-consistency loss that requires a translated image to be mapped back to the original. It is strictly pairwise: every pair of domains needs its own pair of generators and there is no shared embedding of domains. StarGAN [16] showed that a single generator can serve several domains if it is conditioned on a target domain label. The label is a discrete one-hot vector, and a domain classifier in the discriminator enforces that the output belongs to the requested domain. The domains therefore have to be fixed and labeled in advance. StarGAN v2 [17] replaces the label by a style code that is produced by a mapping network or by a style encoder and injected into the generator through adaptive instance normalization, which gives diverse outputs for the same target domain. It was evaluated on human and animal face datasets with visual-realism metrics. Its style codes are still learned per labeled domain, and the evaluation does not test whether translated images help a downstream task on satellite data. StarGAN v2 is the architectural base of the generator in this project, which keeps the idea of injecting a continuous code through AdaIN [26] but obtains the code from a self-supervised encoder instead of labeled domains.

2.1.4 Label-free and multi-domain translation for satellite imagery

Tasar et al. [19] propose StandardGAN for multi-source domain adaptation in the semantic segmentation of VHR satellite images. A GAN standardizes the data distribution of each source and target domain toward a common form, and a single classifier is then trained on the standardized data. The method targets satellite imagery, but it normalizes distributions instead of learning a continuous embedding, it handles one target at a time rather than arbitrary many-to-many translation, and it does not address unseen combinations of domains.

Zhang et al. [18] propose Self-Supervised Domain-Agnostic Domain Adaptation for satellite images, abbreviated SS(DA)^2, which is the base paper of this project. It is a contrastive self-supervised GAN that translates between two satellite domains using only unlabeled imagery and without predefined domain labels. This removes

the label requirement, but the formulation is restricted to two domains, so the method has neither an N-domain mechanism nor a mechanism for unseen pairs. The present work keeps the label-free training signal and removes the two-domain restriction.

Two recent diffusion-based works are also relevant. Vinholi et al. [20] use a diffusion model (DDIM sampling) to translate low-resolution imagery into the high-resolution equivalent of another optical sensor, and they test the result on change detection. The approach is computationally heavy and is trained per pair of sensors, and it includes no test of generalization to unseen sensor pairs. Kieu et al. [21] train a single diffusion noise predictor that is conditioned on domain labels and route translations between domains through a central domain, using $K-1$ paired datasets for K domains. The work is general-purpose and evaluated on natural images, and it still needs every domain to be paired with the central domain, so it supports the observation that the unseen-combination problem is open but does not solve it for satellite data.

2.1.5 Datasets

The experiments of this project need imagery of the same location from several sensors, ideally with co-registered references. PASTIS [27] is a benchmark for the panoptic segmentation of agricultural parcels from Sentinel-2 time series. It was extended with Sentinel-1 radar series (PASTIS-R [28]) and with a very-high-resolution SPOT 6/7 image for each patch (PASTIS-HD [4]), which yields co-registered optical, radar and VHR imagery for 2,433 patches over France, together with crop-type annotations. MSC-France, introduced together with the X-STARS method [29], provides triplets of Sentinel-2, Landsat-8 and SPOT-6 imagery over French cities without labels, which makes it suitable for self-supervised pretraining. MultiEarth 2022 [30] provides a four-sensor time series (Sentinel-1, Sentinel-2, Landsat-5, Landsat-8) over the Amazon with deforestation labels, and SpaceNet 6 [31] pairs SAR and optical imagery over one city with building footprints. These are dataset and challenge papers and do not propose adaptation methods. Their role in this project, as well as the datasets that were actually used in this report, is described in Section 3.3.

2.2 Literature Collection & Segregation

Table 2.1 lists the collected works by category, together with their role in the project. The categories follow the structure of the research problem: the surveys define the problem, the self-supervised works motivate Objective 1, the translation works motivate Objective 2, the satellite-specific translation works define the gap, and the dataset papers provide the data.

TABLE 2.1: Segregation of the collected literature.

Category	**Works**	**Role in the project**
A. Domain adaptation surveys	Tuia et al. (2016) [6]; Xu et al. (2022) [7]; Khazri Khelif et al. (2024) [11]	Problem definition and taxonomy
B. Self-supervised representation learning	SeCo [12]; GASSL [13]; SatMAE [14]; Stojnić and Risojević [15]	Motivation for Objective 1
C. GAN-based translation	CycleGAN [9]; StarGAN [16]; StarGAN v2 [17]	Motivation and architectural base for Objective 2
D. Label-free and multi-domain translation for satellite images	SS(DA)^2 [18]; StandardGAN [19]; Vinholi et al. [20]; Kieu et al. [21]	Definition of the research gap
E. Datasets	PASTIS family [4, 27, 28]; MSC-France [29]; MultiEarth 2022 [30]; SpaceNet 6 [31]	Training and evaluation data

In addition to these works, a set of technical references supports individual design decisions of the framework, such as adaptive instance normalization, the InfoNCE objective, channel attention, spectral normalization and the least-squares GAN loss. They are cited where the corresponding component is described in Chapter 3.

2.3 Summary of Literature

Table 2.2 compares the translation and adaptation methods on the four properties that define the target of this project. The properties are: no explicit domain labels needed for training, a single model for many domains, a continuous representation

of the domain, and a design for satellite imagery. The assessment of each method is taken from the descriptions in the cited papers and is qualitative; it does not compare reported accuracies, because the methods were evaluated on different datasets and tasks.

TABLE 2.2: Comparison of adaptation and translation methods on the properties targeted in this project.

Method	No domain labels	Many domains, one model	Continuous domain code	Satellite imagery
CycleGAN [9]	×	×	×	×
StarGAN [16]	×	✓	×	×
StarGAN v2 [17]	×	✓	✓	×
StandardGAN [19]	×	×	×	✓
SS(DA)^2 [18]	✓	×	×	✓
Vinholi et al. [20]	×	×	×	✓
Kieu et al. [21]	×	✓	×	×
Proposed (Objectives 1–2)	✓	✓	✓	✓

Three conclusions follow from the review. First, self-supervised pretraining is well established for remote sensing, but it has been applied within single sensors and evaluated on classification. Second, image translation across many domains with a single generator is well established for natural images, but it relies on discrete, labeled domains. Third, the translation methods that were designed for satellite imagery either are restricted to two domains or handle one target at a time, or depend on domain-specific training and on heavy iterative sampling. The last row of Table 2.2 states the target of this work; it should be read as a design goal, and Chapter 4 reports how far the implementation achieves it.

2.4 Identification of Gaps

The following gaps are derived from the literature review and are addressed by the two objectives of this report.

G1. No cross-sensor continuous embedding. Self-supervised methods for remote sensing [12–15] stay within a single sensor and produce no continuous embedding that places different sensors, such as SAR and optical, in one space, and none has a generative component.

G2. Discrete, labeled domain conditioning. Multi-domain GANs [16, 17] obtain their domain codes from labeled, enumerated domains, so a domain code cannot be obtained for a sensor without labels.

G3. Pairwise or normalization-based satellite adaptation. The satellite-specific methods are either pairwise [18] or normalize distributions for one target at a time [19]; neither offers arbitrary many-to-many translation from a single model.

G4. Computational cost and domain-specific training of diffusion translators. Diffusion-based alternatives require iterative sampling and are trained per sensor pair [20] or need every domain to be paired with a central domain [21].

The literature also shows that adaptation methods are usually judged by visual realism instead of by their effect on a downstream task, and that generalization to unseen sensor combinations is rarely tested. These two gaps are addressed by the later objectives of the project and are therefore outside the scope of this report.

2.5 Need for Research

The number of satellite sensors and the volume of unlabeled imagery grow much faster than the amount of labeled data. A domain adaptation method that needs neither labels nor a fixed list of domains, and that scales to many sensors with one model, would make labeled data from one sensor reusable for others and would lower the cost of monitoring applications such as crop mapping, deforestation monitoring and disaster response. Because SAR sensors observe through clouds and optical sensors provide interpretable spectral information, a common representation across the two is particularly valuable. Establishing whether a continuous, label-free domain embedding and a single conditional generator are feasible for real multi-sensor satellite data is the first step toward that goal, and it is the aim of the work reported here.

2.6 Problem Statement

Deep learning models trained on satellite imagery from one sensor, season or region degrade sharply when applied to another, and existing remedies require labeled target data and a pre-defined domain boundary, neither of which is practical at scale. A method is therefore needed that learns a continuous representation of sensor domain from unlabeled multi-sensor patches, and that uses this representation to translate imagery into the style of an arbitrary target domain with a single model while keeping the geospatial content of the scene unchanged.

2.7 Objectives

The objectives of this report, which are the first two of the four project objectives, are the following.

1. To design and develop a domain encoder that learns continuous-valued domain embeddings directly from unlabeled multi-sensor satellite image patches.
2. To develop a conditional generator that translates image content into the style of an arbitrary target domain using a single unified model.

Table 2.3 shows where each objective is addressed in this report, so that the conclusions in Chapter 5 can be traced back to the objectives.

Table 2.3: Mapping of the objectives to the sections of this report.

Objective	**Design and method**	**Results**	**Conclusion**
Objective 1 (domain encoder)	Sections 3.4.1 and 3.4.2	Section 4.5	Section 5.2.1
Objective 2 (conditional generator)	Sections 3.4.3, 3.4.4 and 3.4.5	Section 4.6	Section 5.2.2

2.8 Limitations of the Study

This is an academic project with limits on time and compute, and the following limitations should be kept in mind when reading the report. The scope is restricted to the first two objectives. Generalization to sensor pairs that were never seen together during training, and the effect of the translation on a downstream classification or segmentation task, are not evaluated here, and the claims in this report are limited accordingly. The experiments use one real benchmark, PASTIS-HD [4, 27], with three sensor domains over agricultural areas of France, and only 1,500 of its 2,433 patches were converted. The large-scale unlabeled MSC-France corpus [29], which the project plan foresees for encoder pretraining, was not used. All training was done on a CPU-only laptop, which forced a small network width and a short training schedule (Section 3.5). Finally, the modalities in PASTIS-HD are spatially co-registered but were not acquired at the same time, which limits how strictly paired image metrics can be interpreted (Section 3.3).

Chapter 3

PROPOSED METHODOLOGY

This chapter describes how the research was carried out. The methodology follows an iterative design-and-evaluate process. A literature review and gap analysis (Chapter 2) defined the problem; a real multi-sensor benchmark was selected and converted into a unified format; a two-stage framework, consisting of a self-supervised domain encoder followed by a conditional generator, was designed and implemented; the implementation was verified with an automated test suite; and the trained models were evaluated on held-out data with a protocol that is described in Section 3.4.7. Figure 3.1 shows the workflow.

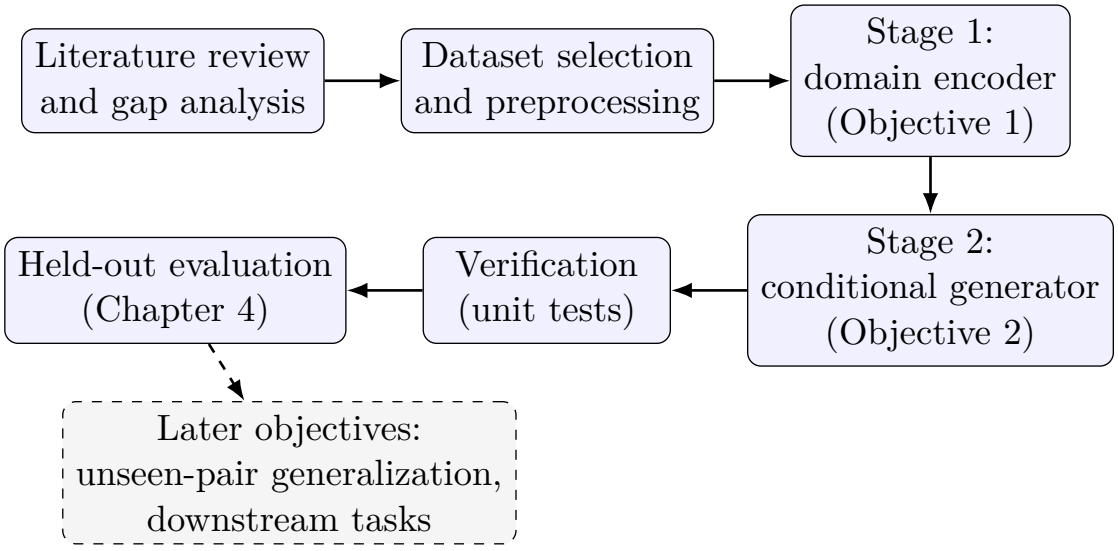


FIGURE 3.1: Workflow of the research methodology. The dashed box is outside the scope of this report.

3.1 Problem formulation

Let $\mathcal{D}_1, \ldots, \mathcal{D}_K$ denote K sensor domains. Each domain produces image patches $x \in \mathbb{R}^{C \times H \times W}$; in this work $C = 4$ and $H = W = 128$. Conceptually, a patch is a rendering of a scene by a sensor, so two patches of the same location seen by different sensors share content (field boundaries, roads, water bodies) but differ in style (radiometry, texture, noise). Training uses only the unlabeled union of patches $\mathcal{U} = \{x_i\}$. The index k of the domain of a patch is not available to the learning algorithm; it is used only afterwards, for evaluation.

The problem is split into two learning tasks that correspond to the two objectives.

Objective 1. Learn an encoder $E_\theta : \mathbb{R}^{C \times H \times W} \to \mathbb{S}^{d-1}$ that maps a patch to a unit-norm embedding $z = E_\theta(x) \in \mathbb{R}^d$, with $d = 64$, such that patches that share sensor characteristics are mapped to nearby points, and such that the encoder can be applied to any patch, including patches of a sensor that was not enumerated during training.

Objective 2. Learn a single generator $G_\psi(x, z)$ such that, for a source patch x and a style embedding $z = E_\theta(x')$ computed from any patch x' of the target domain, the output $\hat{x} = G_\psi(x, z)$ satisfies three requirements: (a) style, $E_\theta(\hat{x}) \approx z$; (b) content, the geometry of x is preserved in $\hat{x}$; and (c) realism, $\hat{x}$ is indistinguishable from real patches of the target domain. Because the target is specified only through z, one generator represents all $K(K-1)$ ordered translations and the K identity mappings.

3.2 Proposed research framework

The proposed framework has two training stages and one inference procedure, shown in Figure 3.2.

Stage 1 (Objective 1). The domain encoder is trained on unlabeled patches from all sensors with a contrastive objective (Section 3.4.2). No domain label is used.

Stage 2 (Objective 2). The encoder is frozen. The generator and a discriminator are trained adversarially, with a source patch from any domain and a target

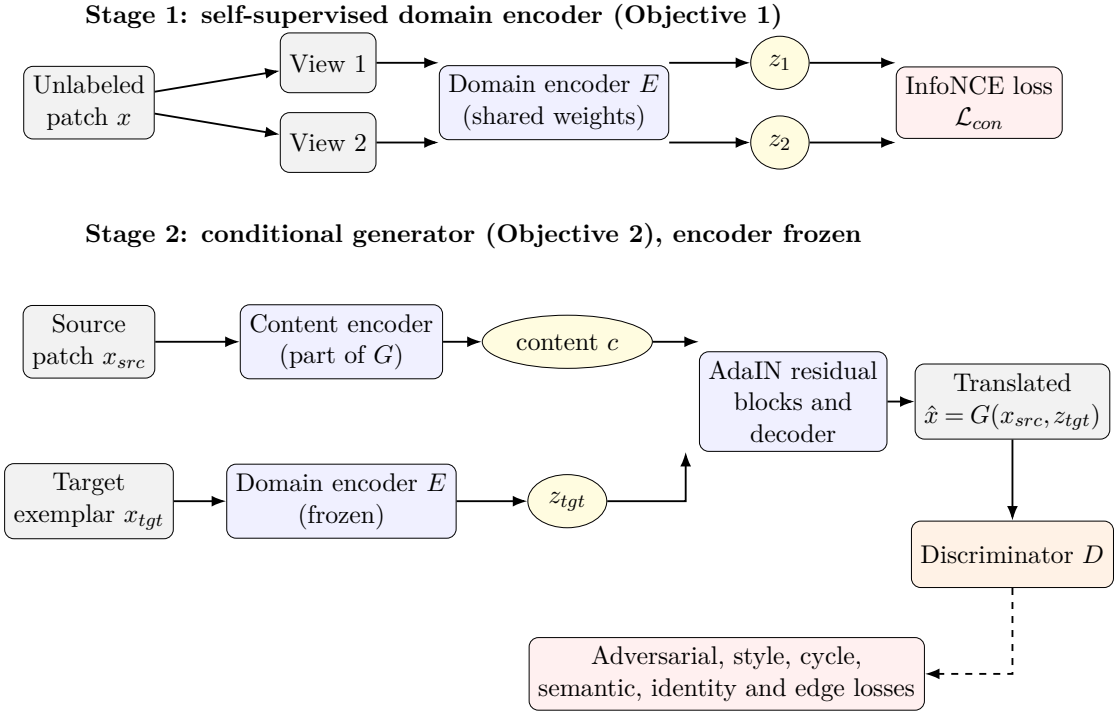


FIGURE 3.2: Two-stage framework. In Stage 1 the encoder is trained with a contrastive objective on two augmented views of unlabeled patches. In Stage 2 the encoder is frozen; it supplies the style embedding z_{tgt} of a target exemplar to the generator, which combines it with the content of a source patch.

exemplar from a randomly chosen domain. The style vector of the target exemplar, computed by the frozen encoder, conditions the generator.

Inference. To translate a patch into a target style, the style vector is taken from a single target exemplar, from the normalized mean of the embeddings of many target patches (a domain centroid), or from a convex combination of two such vectors (interpolation). A single forward pass of the generator produces the result.

Table 3.1 summarizes how the components correspond to the objectives.

3.3 Data collection and preprocessing

3.3.1 Datasets in the project plan and datasets used

The project plan assigns datasets to the objectives as shown in Table 3.2. This report uses PASTIS-HD only. MSC-France, which the plan names as the primary

TABLE 3.1: Framework components and the objectives they serve.

Component	Function	Objective
Domain encoder E	Continuous, label-free embedding of patch style	Objective 1
Contrastive loss	Self-supervised training signal for E	Objective 1
Generator G	Single model for all target styles	Objective 2
Discriminator D	Realism and style-consistency of outputs	Objective 2
Composite generator loss	Style transfer with content preservation	Objective 2

corpus for encoder pretraining, and MultiEarth 2022 were not used in the experiments; a converter for MSC-France triplets exists in the code base, but the dataset was not downloaded. The consequences of this are discussed in Section 4.8.

TABLE 3.2: Datasets in the project plan and their status in this report.

Dataset	Planned role	Status in this report
PASTIS-HD [4, 27, 28]	Primary for the generator; secondary (unlabeled) for the encoder	Used for both stages and for the evaluation
MSC-France [29]	Primary for encoder pretraining (unlabeled, about 4,500 triplets of Sentinel-2, Landsat-8, SPOT-6)	Not used
MultiEarth 2022 [30]	Secondary for the generator; four sensors	Not used

3.3.2 PASTIS-HD

PASTIS-HD contains 2,433 patches of 128×128 pixels over four regions of metropolitan France, covering about 4,000 km^2 [4, 27]. For each patch it provides a Sentinel-2 time series with 10 spectral bands at 10 m and 38 to 61 acquisitions, Sentinel-1 series in ascending and descending orbits with about 70 acquisitions each, and a single SPOT 6/7 image with 3 bands that is natively at 1.5 m, resampled to 1 m (1280×1280 pixels) and aligned pixel by pixel with the Sentinel data. Crop-type

annotations of 18 classes are also included; they are not used in the two objectives of this report.

3.3.3 Conversion to a unified multi-sensor format

A conversion script matches the patches of the three modalities by their common patch identifier and writes one tensor per modality and location. All tensors have 4 channels and 128×128 pixels and are scaled to $[-1, 1]$, so that a single encoder and generator can process every domain. The domains and their channels are the following.

- **Sentinel-2.** The first temporal acquisition is used. The bands B4, B3, B2 and B8 (red, green, blue, near-infrared) are selected and scaled as $x' = 2\,\mathrm{clip}(x/5000, 0, 1) - 1$.
- **SPOT 6/7.** The image is resized from 1280×1280 to 128×128 by bilinear interpolation, and the three optical bands are scaled by $x' = 2x/255 - 1$. The fourth channel is not a measured band: it is a proxy computed as the mean of the red and green bands.
- **Sentinel-1 SAR.** The first ascending acquisition is used. The channels are the VV and VH backscatter, their difference, and their mean, all interpreted as decibel values, clipped to $[-30, 0]$ and scaled linearly to $[-1, 1]$.

The SPOT proxy channel and the clipping of the SAR difference channel are limitations of the preprocessing. Because the difference between VV and VH in decibels is mostly positive, clipping at 0 dB saturates this channel at $+1$ for most pixels; the measured fraction of saturated values is reported in Section 4.4.

3.3.4 Subset, split and temporal alignment

The converter processes the aligned patches in sorted identifier order and was run with a limit of 1,500 patches, so the first 1,500 of the 2,433 patches are used for each of the three domains. These patches come from three Sentinel-2 tiles: T30UXV (531 patches), T31TFJ (623) and T31TFM (346); the fourth tile of PASTIS-HD is not included. The data are split by sorted index into 80% for training and 20%

for held-out evaluation, which gives 1,200 training and 300 held-out patches per domain. Because the split follows the identifier order, the 300 held-out patches all belong to tile T31TFM, and 46 further T31TFM patches are in the training set. The held-out evaluation is therefore mainly a test on a different region from most of the training data.

The three modalities are co-registered in space, but they were not acquired at the same time. For the selected patches, the first Sentinel-2 acquisition is between 17 and 24 September 2018 and the first Sentinel-1 acquisition between 1 and 4 October 2018, so that the two differ by 11 days at the median (at most 14 days). The SPOT imagery comes from 2019 (the source file names contain the year 2019), which is between about three and fifteen months after the Sentinel acquisitions. Vegetation and crop state can therefore differ between a patch and its "co-registered" counterpart in another domain.

3.3.5 Augmentation and synthetic data

During training, patches are augmented by random horizontal and vertical flips and random rotations by multiples of 90°; in Stage 2, small Gaussian noise (standard deviation drawn uniformly from $[0, 0.01]$) is also added to the source patch. The held-out patches are not augmented. For software development and testing, the code base also contains a generator of synthetic multi-sensor scenes with four simulated sensor domains and four simulated viewing-angle domains; it is not used for any result in this report.

3.4 Algorithm & Model design

3.4.1 Domain encoder

The encoder E_θ (888,224 trainable parameters) consists of the following parts; the layer-wise specification is given in Appendix A.

1. An input-adapter convolution (3×3, 4 to 32 channels), so that the same encoder can process optical, VHR or SAR channels.

2. Three stride-2 convolutional blocks (4×4 kernels; 32 to 64, 64 to 128 and 128 to 256 channels), each followed by instance normalization and a LeakyReLU activation, which reduce the resolution from 128×128 to 16×16.

3. A squeeze-and-excitation (SE) block [32] that re-weights the 256 channels with a gate computed from the globally averaged features, so that the encoder can emphasize the most discriminative bands.

4. Global first- and second-order statistics: the channel-wise mean and standard deviation of the final feature map,

$$\mu_c = \frac{1}{HW} \sum_{i,j} h_{c,i,j}, \qquad \sigma_c = \sqrt{\frac{1}{HW} \sum_{i,j} (h_{c,i,j} - \mu_c)^2 + \epsilon}, \tag{3.1}$$

   where h is the feature map. Mean and standard deviation summarize radiometry and texture independently of the spatial arrangement, which is why they are also the quantities used by instance normalization for style [26, 33].

5. A multilayer perceptron head ($544 \to 256 \to 128 \to 64$ with LeakyReLU and dropout of 0.1), whose 544 inputs are the 512 statistics and a 32-dimensional angle feature that is zero in this work, followed by L2 normalization onto the unit hypersphere,

$$z = \frac{f_\phi([\mu;\sigma])}{\left\| f_\phi([\mu;\sigma]) \right\|_2}, \qquad z \in \mathbb{R}^{64}. \tag{3.2}$$

The implementation also has an optional branch that fuses a viewing-angle feature with the image statistics, for angle domains. It receives a neutral zero input when no angle is given, which is the case for all PASTIS-HD experiments.

3.4.2 Contrastive pretext task

The encoder is trained without domain labels by an instance-discrimination task in the style of InfoNCE [22, 24]. Two independently augmented views of the same patch form a positive pair, and the other patches of the mini-batch are negatives. With $z_{1,i}$ and $z_{2,i}$ denoting the embeddings of the two views of patch i, a batch of size B, and a temperature τ, the symmetric loss is

$$\mathcal{L}_{con} = -\frac{1}{2B} \sum_{i=1}^{B} \left[\log \frac{e^{z_{1,i} \cdot z_{2,i}/\tau}}{\sum_j e^{z_{1,i} \cdot z_{2,j}/\tau}} + \log \frac{e^{z_{2,i} \cdot z_{1,i}/\tau}}{\sum_j e^{z_{2,i} \cdot z_{1,j}/\tau}} \right]. \tag{3.3}$$

Since embeddings are unit vectors, the dot product is the cosine similarity. The loss is minimized when the two views of a patch are closer to each other than to any other patch of the batch. It follows that the signal that the encoder receives is patch-level discriminability that is invariant to flips and rotations; the sensor identity is not supervised, and the extent to which it is captured emerges from the data and is measured in Section 4.5. Algorithm 1 summarizes Stage 1.

Algorithm 1 Stage 1: self-supervised training of the domain encoder

Require: Unlabeled multi-sensor patches $\mathcal{U}$, temperature τ, epochs T, batch size B

for epoch $= 1$ **to** T **do**
  for each mini-batch $\{x_i\}_{i=1}^{B}$ drawn from $\mathcal{U}$ **do**
    $v_1 \leftarrow \text{augment}(x)$, $v_2 \leftarrow \text{augment}(x)$ (independent flips and rotations)
    $z_1 \leftarrow E_\theta(v_1)$, $z_2 \leftarrow E_\theta(v_2)$
    compute $\mathcal{L}_{con}$ by Equation 3.3
    update θ with AdamW
return E_θ

3.4.3 Unified conditional generator

The generator G_ψ (1,594,116 trainable parameters) has an encoder–bottleneck–decoder structure in the spirit of StarGAN v2 [17], adapted so that it needs no domain labels. The separation of a content code from a style code follows the content–style decomposition used in multimodal image translation [34].

1. A *content encoder* with a reflection-padded 7×7 convolution and two stride-2 convolutional blocks, with instance normalization, reduces x_{src} (128×128, 4 channels) to a content code $c = \text{Enc}_c(x_{src})$ with 128 channels at 32×32.

2. Four *residual blocks* with adaptive instance normalization (AdaIN) [26] process the content code. In each block a style-dependent scale and shift are applied after normalizing each channel,

$$\text{AdaIN}(h, z) = \gamma(z) \cdot \frac{h - \mu(h)}{\sigma(h)} + \beta(z), \qquad \gamma(z) = W_\gamma z + b_\gamma, \quad \beta(z) = W_\beta z + b_\beta, \tag{3.4}$$

   where $\mu(h)$ and $\sigma(h)$ are the per-channel statistics of the feature map h. The parameters γ and β are computed from the target embedding z_{tgt} by learned linear layers, so that the target style enters through the statistics of the features, at every block.

3. A *decoder* with two bilinear $2\times$ upsampling steps, each followed by a convolution and an AdaIN layer, and a final reflection-padded 7×7 convolution with a tanh activation. Bilinear upsampling is used instead of transposed convolution to avoid checkerboard artifacts.

The output is $\hat{x} = G_\psi(x_{src}, z_{tgt})$. The only place where the target domain is specified is z_{tgt}, so the same weights serve all targets.

3.4.4 Discriminator with style projection

The discriminator D_ξ (709,601 trainable parameters) is a patch-based convolutional network [35]. Spectral normalization [36] is applied to every convolution to keep training stable. Besides an unconditional real/fake output, it computes a projection term between its features and the conditioning embedding, in the manner of projection discriminators [37],

$$D_\xi(x, z) = D_{uncond}(x) + \frac{\langle \phi(x), W_{proj}\, z \rangle}{\sqrt{d_\phi}}, \tag{3.5}$$

where $\phi(x)$ is the feature map of the last convolutional block, W_{proj} a learned linear map and $d_\phi = 256$ the feature dimension. The term rewards outputs whose appearance is consistent with the style vector.

3.4.5 Composite training objective

The discriminator is trained with the least-squares GAN loss [38], and the generator with a weighted sum of six terms. Let x_{tgt} be a target exemplar, $z_{src} = E(x_{src})$,

$z_{tgt} = E(x_{tgt})$, and $\hat{x} = G(x_{src}, z_{tgt})$.

$$\mathcal{L}_D = \tfrac{1}{2}\mathbb{E}\Big[(D(x_{tgt}, z_{tgt}) - 1)^2\Big] + \tfrac{1}{2}\mathbb{E}\Big[D(\hat{x}, z_{tgt})^2\Big] \tag{3.6}$$

$$\mathcal{L}_{adv} = \mathbb{E}\Big[(D(\hat{x}, z_{tgt}) - 1)^2\Big] \tag{3.7}$$

$$\mathcal{L}_{sty} = \mathbb{E}\,\|E(\hat{x}) - z_{tgt}\|_1 \tag{3.8}$$

$$\mathcal{L}_{cyc} = \mathbb{E}\,\|G(\hat{x}, z_{src}) - x_{src}\|_1 \tag{3.9}$$

$$\mathcal{L}_{sem} = \mathbb{E}\,\|\mathrm{Enc}_c(\hat{x}) - \mathrm{Enc}_c(x_{src})\|_1 \tag{3.10}$$

$$\mathcal{L}_{id} = \mathbb{E}\,\|G(x_{src}, z_{src}) - x_{src}\|_1 \tag{3.11}$$

$$\mathcal{L}_{edge} = \mathbb{E}\,\|\nabla_S(\hat{x}) - \nabla_S(x_{src})\|_1 \tag{3.12}$$

The adversarial loss (Equation 3.7) makes $\hat{x}$ resemble real patches of the target style. The style-reconstruction loss (Equation 3.8) requires the target style to be measurable in the output by re-encoding it. The cycle-consistency loss [9] (Equation 3.9) requires that translating back to the source style recovers the source patch, which discourages the generator from discarding content. The semantic-preservation loss (Equation 3.10) penalizes changes in the content code, and the edge-preservation loss (Equation 3.12), in which ∇_S is the Sobel gradient magnitude, keeps parcel boundaries and road edges. The identity loss (Equation 3.11) requires that conditioning on the patch's own style leaves it unchanged. The total generator loss is

$$\begin{aligned}\mathcal{L}_G = {} & \lambda_{adv}\mathcal{L}_{adv} + \lambda_{sty}\mathcal{L}_{sty} + \lambda_{cyc}\mathcal{L}_{cyc} \\ & + \lambda_{sem}\mathcal{L}_{sem} + \lambda_{id}\mathcal{L}_{id} + \lambda_{edge}\mathcal{L}_{edge}.\end{aligned} \tag{3.13}$$

The framework also defines a geometric-preservation loss $\mathcal{L}_{geo}$, an SSIM-style comparison of content features that is active only when the source or target is a viewing-angle domain. As the PASTIS-HD experiments contain only sensor domains, it is inactive here.

3.4.6 Training procedure and hyperparameters

Stage 2 is summarized in Algorithm 2, and the hyperparameters of both stages are listed in Table 3.3. In every iteration a source patch is drawn from the training set, a target domain is drawn uniformly at random (it may equal the source domain),

and a target exemplar is drawn at random from that domain. The exemplar is not the co-registered patch of the source location, so training is unpaired.

Algorithm 2 Stage 2: training of the generator and the discriminator

Require: Frozen encoder E, training patches, epochs T, loss weights λ_*

for epoch $= 1$ **to** T **do**
  for each mini-batch **do**
    draw source patches x_{src}; draw a random target domain and target exemplars x_{tgt}
    $z_{src} \leftarrow E(x_{src}), \quad z_{tgt} \leftarrow E(x_{tgt})$
    $\hat{x} \leftarrow G(x_{src}, z_{tgt})$
    update D by minimizing $\mathcal{L}_D$ (Equation 3.6) with $\hat{x}$ detached
    compute $\mathcal{L}_{adv}$, $\mathcal{L}_{sty}$, $\mathcal{L}_{cyc}$, $\mathcal{L}_{sem}$, $\mathcal{L}_{id}$, $\mathcal{L}_{edge}$
    update G by minimizing $\mathcal{L}_G$ (Equation 3.13)
  step the learning-rate schedules
return G, D

TABLE 3.3: Hyperparameters of the two training stages.

Stage 1: domain encoder	**Value**
Embedding dimension d	64
Optimizer	AdamW [39]
Learning rate; weight decay	5×10^{-4}; 1×10^{-4}
Temperature τ	0.07
Batch size; epochs	8; 10 (4,500 iterations)
Stage 2: generator and discriminator	**Value**
Base channel width; residual blocks	32; 4
Optimizer (both networks)	Adam [40], $\beta_1 = 0.5$, $\beta_2 = 0.999$
Learning rate	3×10^{-4} with cosine annealing [41] to 10^{-5}
Gradient clipping (maximum norm)	1.0
Batch size; epochs	8; 12 (5,400 iterations)
λ_{adv}, λ_{sty}, λ_{id}	1.0 each
λ_{sem}, λ_{edge}	2.0 each
λ_{cyc}	10.0

3.4.7 Evaluation protocol

The models are evaluated on the 300 held-out patches per domain, with a fixed random seed. Domain labels and the co-registered patches are used only for evaluation and never for training. The style of a domain is represented by its *centroid*, the L2-normalized mean of the embeddings of the 1,200 training patches of that domain.

Objective 1 (encoder). (i) Domain decodability: the accuracy of assigning each held-out embedding to the nearest domain centroid, and the accuracy of a 5-nearest-neighbor classifier that uses the training embeddings (chance level is 1/3). (ii) Cluster quality: the mean silhouette coefficient with cosine distance, the mean cosine similarity within and between domains, and the fraction of the total embedding variance that lies between domains. (iii) Effective dimensionality: the participation ratio $(\sum_i \lambda_i)^2 / \sum_i \lambda_i^2$ of the covariance eigenvalues λ_i of the embeddings. (iv) Pretext performance on held-out data: the top-1 retrieval accuracy of the second view among all 900 held-out patches, the InfoNCE loss for batches of 8 (the value for chance-level embeddings is $\ln 8 = 2.08$), and the mean cosine similarity of positive and of negative pairs. (v) Cross-sensor retrieval of the same location, to measure how much scene content the embedding carries.

Objective 2 (generator). For each ordered pair of source and target domains, all held-out patches are translated with the target centroid as style. The output is compared with the co-registered real target patch by the mean absolute error (L1), the peak signal-to-noise ratio (PSNR) and the structural similarity index (SSIM, Gaussian window of 11×11 with $\sigma = 1.5$), all computed on images rescaled to $[0, 1]$. Two reference baselines are used: the untranslated source patch, and a constant image equal to the mean of the training patches of the target domain. As a control, the style of the third domain is used instead of the target style. Further measurements are the fraction of translated outputs whose re-encoded embedding is nearest to the target centroid (style transfer success), the Pearson correlation between the Sobel edge maps of the source and the output (content preservation), the PSNR and SSIM of the cycle reconstruction $G(G(x, \bar{z}_{tgt}), \bar{z}_{src})$, and the reconstruction quality when the generator is conditioned on the patch's own embedding. Continuity of the style space is examined by interpolating between two centroids, $z(\alpha) = (1-\alpha)\bar{z}_a + \alpha\bar{z}_b$ (renormalized to unit length) with $\alpha \in \{0, 0.25, 0.5, 0.75, 1\}$, and measuring how the output changes with α.

3.5 System requirements

The implementation is written in Python with PyTorch [42]. Table 3.4 lists the environment in which the models were developed, trained and evaluated. The GPU listed was not used, because the installed PyTorch build is the CPU-only build; all training and evaluation ran on the CPU.

Table 3.4: Hardware and software environment.

Item	**Specification**
Processor	Intel Core i7-9750H, 2.60 GHz, 6 cores / 12 threads
Memory	7.8 GB RAM
Graphics	NVIDIA GeForce GTX 1650 (not used, CPU-only PyTorch build)
Operating system	Microsoft Windows 11 Home
Language and framework	Python 3.13.2, PyTorch 2.11.0 (CPU)
Libraries	NumPy 2.2.3, SciPy 1.16.1, Matplotlib 3.10.3, Pillow 11.3.0, PyYAML, tqdm
Storage	About 1.7 GB for the converted PASTIS-HD subset; model checkpoints of 3.6 MB (encoder), 6.4 MB (generator) and 2.9 MB (discriminator)

The code base contains an automated test suite of 29 tests for the configuration, the dataset, the encoder, the losses, the metrics and the model shapes; all 29 tests pass in the environment above. Training is started by a single command-line entry point that runs the data conversion, Stage 1 and Stage 2 in sequence. The configuration defines a seed of 42, but the implementation uses it only for the synthetic-data generator and does not seed the random number generators of the training loops, so re-running the training will not reproduce identical weights. The evaluation of Chapter 4, in contrast, is deterministic given the trained checkpoints, because its script fixes all random choices, and it can be repeated as described in Appendix B.

Chapter 4

RESULTS AND DISCUSSIONS

This chapter summarizes what the literature review established, summarizes the proposed design, describes how it was implemented and trained, and reports and discusses the results for the two objectives. All quantitative results in Sections 4.4 to 4.6 were produced by the evaluation script described in Section 3.4.7 on the 300 held-out patches per domain, using the stored checkpoints; they are single measurements of one trained model, not averages over repeated training runs.

4.1 Major findings from Literature

The literature review of Chapter 2 led to the following findings, which shaped the design.

F1. Domain adaptation for remote sensing is well surveyed, and the multi-domain setting is defined conceptually [7], but no surveyed method offers a single model that adapts many sensors without domain labels [6, 7, 11].

F2. Self-supervised pretraining, in particular contrastive learning with the InfoNCE objective, is effective for remote sensing imagery, but has been applied within one sensor or sensor family [12–15].

F3. Translation between many domains with one generator is mature for natural images, and a continuous style code injected by adaptive instance normalization gives diverse outputs [16, 17]; however, the codes are derived from labeled, enumerated domains.

F4. The satellite-specific translators are pairwise [18] or normalize distributions for one target at a time [19].

F5. Diffusion-based translators achieve good fidelity but need iterative sampling and either per-pair training or a central domain [20, 21].

F6. Adversarial training combined with cycle consistency and explicit content constraints is the standard recipe for translation that preserves content [9, 10], and co-registered multi-sensor datasets such as PASTIS-HD make an evaluation against real target-sensor patches possible [4].

4.2 Summary of the proposed design

The proposed design combines a self-supervised, contrastively trained domain encoder with a single AdaIN-conditioned generator, as described in Chapter 3. Table 4.1 lists the main design decisions and the reasons for them.

4.3 Implementation and experimental setup

The two stages were implemented and trained as described in Section 3.4.6, on the 3,600 training patches (1,200 per domain) of the PASTIS-HD subset. The checkpoints that are evaluated here are the encoder, generator and discriminator files in the output directory `outputs_pastis_fixed`. Their tensor shapes confirm the architecture in Appendix A: a base channel width of 32, an embedding dimension of 64 and four residual blocks. The encoder has 888,224 and the generator 1,594,116 trainable parameters, and the training loops themselves were not logged (Appendix B), so learning curves are not shown. The evaluation script fixes every random choice, so the numbers below can be regenerated exactly from the checkpoints.

4.4 Characteristics of the converted data

Table 4.2 gives the per-channel statistics of the held-out patches after preprocessing, together with the fraction of values that lie at the limits of the $[-1, 1]$

Table 4.1: Main design decisions and their rationale.

Design decision	**Rationale**
Unit-norm embedding, $d = 64$	Cosine geometry suits the contrastive objective; embeddings are comparable and can be interpolated by a renormalized convex combination.
Channel-wise mean and standard deviation pooling	Summarizes radiometry and texture independently of spatial layout, as the statistics that carry style in instance normalization.
Squeeze-and-excitation attention	Lets the encoder weight the most discriminative bands of each sensor.
Instance-discrimination contrastive loss	Gives a training signal without any domain label.
Single AdaIN-conditioned generator	One model for all ordered domain pairs; the target style enters through feature statistics at every residual block.
Content encoder inside the generator	Provides an explicit content code, so that content drift can be penalized.
Bilinear upsampling in the decoder	Avoids checkerboard artifacts of transposed convolutions.
Projection discriminator, spectral normalization, least-squares loss	Stable adversarial training that is conditioned on the style embedding.
Six-term generator loss	Combines style transfer with cycle, semantic, identity and edge constraints that preserve geospatial content.

range. Two observations matter for the interpretation of the later results. First, the Sentinel-2 patches are dark on this scale (the mean of the red, green and blue channels is between -0.62 and -0.77, which corresponds to 0.11 to 0.19 on a $[0, 1]$ scale), because the reflectance values are divided by a fixed constant of 5000. The SPOT patches are considerably brighter. Second, the SAR difference channel is effectively constant: 97.9% of its values are saturated at $+1$, and its standard deviation is 0.017. The SAR domain therefore carries three informative channels instead of four. The saturated channel is also the reason why SAR patches are displayed in a blue-violet tone in the figures of this chapter, because the display shows the first three channels (VV, VH and the difference) as red, green and blue.

TABLE 4.2: Statistics of the held-out patches after preprocessing (300 patches per domain, values in $[-1, 1]$).

Domain	Channel	Mean	Std	% at +1	% at −1
Sentinel-2	Red	−0.62	0.25	0.35	0.21
	Green	−0.68	0.18	0.34	0.12
	Blue	−0.77	0.16	0.33	0.31
	Near-infrared	0.02	0.26	0.47	0.00
SPOT 6/7	Red	−0.35	0.32	0.00	0.00
	Green	−0.23	0.22	0.00	0.00
	Blue	−0.48	0.24	0.00	0.00
	Proxy (red/green mean)	−0.29	0.26	0.00	0.00
Sentinel-1 SAR	VV	0.17	0.22	0.13	0.00
	VH	−0.26	0.23	0.01	0.09
	VV−VH difference	1.00	0.02	97.86	0.00
	VV, VH mean	−0.04	0.20	0.02	0.00

4.5 Results for Objective 1: the domain encoder

Table 4.3 summarizes the evaluation of the encoder on the 900 held-out patches (300 per domain), and Figure 4.1 shows the embeddings projected onto their first two principal components.

The pretext task is solved. On held-out patches, the contrastive loss is 0.24 against 2.08 for an uninformative embedding, and the second view of a patch is retrieved among 900 candidates in 88.2% of the cases. The positive-pair cosine similarity of 0.9995 shows that the embedding is practically invariant to flips and rotations, which is expected because the mean and standard deviation pooling discards the spatial arrangement of the features.

The embedding is continuous and not collapsed. The embeddings do not form three tight clusters. The mean cosine similarity within a domain is only 0.32, and Figure 4.1 shows that the patches of each sensor are spread widely along a ring-like structure of the projection. At the same time, the effective dimensionality is low: the participation ratio is 3.85, and two principal components hold 65.8% of the variance.

Sensor information is present but moderate. A patch is assigned to its own sensor by the nearest domain centroid in 65.8% of the cases and by 5-nearest-neighbor voting in 63.7% of the cases, which is well above the chance level of

TABLE 4.3: Evaluation of the domain encoder on held-out patches.

Measurement	**Result**
Pretext task	
InfoNCE loss on held-out batches of 8	0.24 (chance level: 2.08)
Top-1 retrieval of the second view among 900 patches	88.2% (chance level: 0.1%)
Cosine similarity of positive / negative pairs	0.9995 / 0.134
Sensor information in the embedding (chance level: 33.3%)	
Nearest-centroid domain accuracy	65.8%
recall for Sentinel-2 / SPOT 6/7 / SAR	62.7% / 74.3% / 60.3%
5-nearest-neighbor domain accuracy	63.7%
Geometry of the embedding	
Mean cosine similarity within / between domains	0.321 / 0.041
within Sentinel-2 / SPOT 6/7 / SAR	0.461 / 0.204 / 0.298
Mean silhouette coefficient (cosine distance)	0.110
Fraction of variance between domains	21.8%
Participation ratio (effective dimensionality, of 64)	3.85
Variance in principal components 1 and 2	37.8% and 28.0%
Content shared across sensors	
Same-location retrieval across sensors, top-1	0.3% (chance level: 0.3%)
Same-location retrieval across sensors, top-5	1.7% to 2.7% (chance level: 1.7%)

33.3% and well below a clean separation. The confusion matrix in Table 4.4 shows that Sentinel-2 patches are frequently mistaken for SAR patches (82 of 300), which agrees with the positive cosine similarity of 0.15 between the two domain centroids in the same table. The silhouette coefficient of 0.11 and the between-domain share of 21.8% of the variance confirm that the sensors overlap in the embedding space. SPOT patches, which have the largest brightness and contrast, are recognized best (74.3%).

The embedding does not encode content that is shared across sensors. When the embedding of a patch of one sensor is used to retrieve the co-registered patch of another sensor among the 300 held-out locations, the top-1 accuracy is 0.3%, which equals the chance level, and the top-5 accuracy is 1.7% to 2.7%, close to the chance level of 1.7%. The embedding therefore does not describe scene

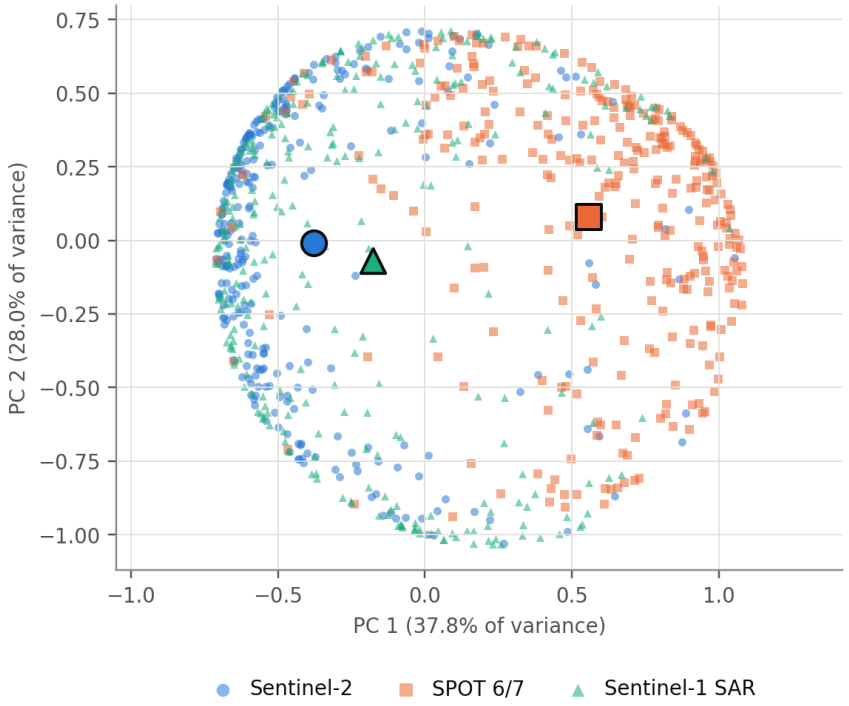


FIGURE 4.1: Held-out embeddings projected onto their first two principal components. Small markers are patches; large outlined markers are the domain means of the projected embeddings.

TABLE 4.4: Nearest-centroid classification of held-out patches (rows: true domain, columns: assigned domain) and cosine similarity between the domain centroids.

	S2	**SPOT**	**SAR**	**Centroid pair**	**Cosine**
Sentinel-2	188	30	82	Sentinel-2, SPOT 6/7	−0.04
SPOT 6/7	61	223	16	Sentinel-2, SAR	0.15
SAR	55	64	181	SPOT 6/7, SAR	−0.03

content in a way that is common to the sensors, which is desirable for a style descriptor, although the different acquisition dates of the modalities (Section 3.3) also reduce such a signal.

4.6 Results for Objective 2: the conditional generator

All translations in this section use the domain centroid of the target as style vector, as described in Section 3.4.7, and the outputs are compared with the co-registered

real patch of the target sensor. Table 4.5 gives the size and speed of the models.

TABLE 4.5: Model size and inference time on the CPU (6 threads, 128×128 patches).

Network	Parameters	One patch (ms)	Per patch, batch of 16 (ms)
Domain encoder E	888,224	15.6	13.2
Generator G	1,594,116	91.7	84.0
Discriminator D (training only)	709,601	–	–

4.6.1 Qualitative results

Figure 4.2 shows, for one held-out patch of each source domain, the translation into each of the two other domains next to the real co-registered patch of that domain. The patches were chosen by a fixed random seed before the results were inspected. The main observations are the following.

- The scene structure is preserved. Field parcels, buildings, the river and the road remain recognizable in the outputs, in particular for the SAR target.
- Translations into SAR are smooth and violet, similar in tone to real SAR patches, but they lack the speckle texture of real SAR imagery.
- Translations into SPOT from a Sentinel-2 source are colored and textured like a SPOT patch, but with a painted look and colors that differ from the real SPOT patch; from a SAR source, the result is a gray and noisy texture with weak structure.
- Translations into Sentinel-2 are the weakest. Both outputs in the figure are a blurred, low-contrast purple-gray haze in which the scene structure is barely visible, and they do not resemble the dark real Sentinel-2 patches.

4.6.2 Quantitative fidelity with respect to the co-registered reference

Table 4.6 compares the translation with three reference images: the untranslated source patch, a constant image equal to the mean of the training patches of the

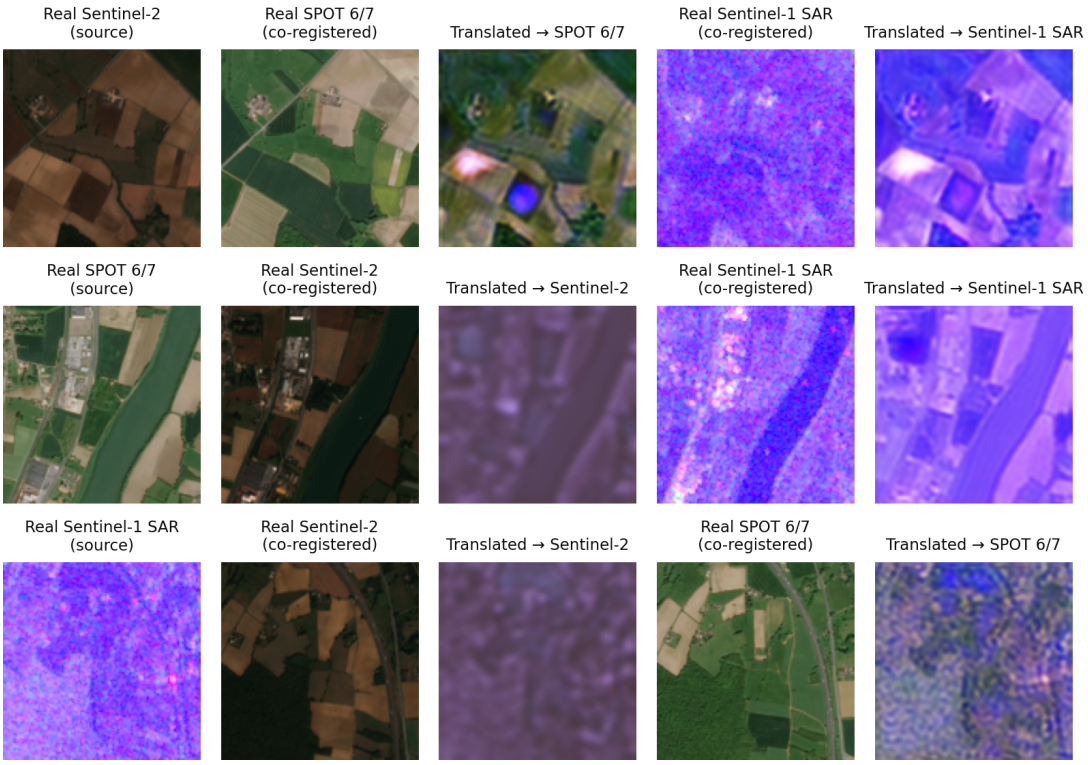


FIGURE 4.2: Translations of one held-out patch per source domain (first column of each row) into the two other domains, next to the real co-registered patch of each target domain. The style is the centroid embedding of the target domain. Sentinel-1 patches are displayed as VV, VH and the saturated difference channel in red, green and blue.

target domain, and the translation obtained with the centroid of the third domain as a wrong style. The standard deviation of the PSNR over the 300 patches is between 0.2 and 1.9 dB for the values in this table.

TABLE 4.6: Agreement with the co-registered real target patch, as PSNR in dB / SSIM (higher is better; mean over 300 held-out patches). S2 denotes Sentinel-2.

Source → target	**Translation**	**Source copied**	**Constant target mean**	**Wrong style**
S2 → SPOT	14.6 / 0.39	13.1 / 0.39	17.2 / 0.48	7.9 / 0.34
S2 → SAR	14.6 / 0.30	5.7 / 0.12	20.5 / 0.44	7.1 / 0.13
SPOT → S2	13.5 / 0.42	13.1 / 0.39	17.8 / 0.50	5.9 / 0.29
SPOT → SAR	16.8 / 0.37	7.4 / 0.17	20.5 / 0.44	9.6 / 0.30
SAR → S2	13.3 / 0.36	5.7 / 0.12	17.8 / 0.50	10.0 / 0.16
SAR → SPOT	13.0 / 0.23	7.4 / 0.17	17.2 / 0.48	15.2 / 0.41

Three findings follow from Table 4.6.

1. **The translation is better than not translating.** The PSNR of the translation is higher than that of the copied source for all six pairs. The gain is largest for pairs that involve SAR (5.7 to 9.4 dB), because the channels of a SAR patch have a different meaning than those of an optical patch, and it is small for the two optical sensors (1.5 dB for Sentinel-2 to SPOT and 0.4 dB for SPOT to Sentinel-2, at an unchanged or slightly higher SSIM).

2. **The style vector matters.** With the correct target style the PSNR is higher than with a wrong style for five of the six pairs, by 3.3 to 7.6 dB. SAR to SPOT is the exception: the Sentinel-2 style gives 15.2 dB against 13.0 dB with the correct SPOT style.

3. **The translation does not reproduce the target radiometry.** A constant image with the mean of the target domain is closer to the real target patch than the translation for all six pairs, by 2.6 to 5.9 dB in PSNR and by 0.07 to 0.25 in SSIM. Pixel-level agreement with the real target is therefore not achieved. The acquisition dates of the sensors differ (Section 3.3), so part of the disagreement is due to real changes on the ground, but this does not explain why the translation is worse than a constant image. Conditioning on the embedding of the co-registered target patch itself instead of the centroid did not help either: it gave a lower PSNR than the centroid for five of the six pairs.

4.6.3 Style transfer and content preservation

Table 4.7 reports whether the outputs are recognized as belonging to the target domain by the encoder, whether the structure of the source is preserved, and how well the source is recovered after a round trip.

- **Style transfer succeeds for some targets and fails for others.** The encoder assigns 80% of the SPOT-to-SAR outputs to SAR, which is more than the 60% of real SAR patches that it assigns correctly. SPOT is reached in 57% (from Sentinel-2) and 45% (from SAR) of the cases, against 74% for real SPOT patches. Translations into Sentinel-2 are assigned to Sentinel-2 in only 29% and 12% of the cases, which is at or below the chance level of 33% and far below the 63% of real patches, in agreement with the haze in Figure 4.2. SAR is also missed when the source is Sentinel-2 (18%).

TABLE 4.7: Style transfer, content preservation and cycle reconstruction on held-out patches. "Real patches assigned" is the share of real target patches that the encoder assigns to their own domain; the value in parentheses is the edge correlation between the real source and the real target patch. S2 denotes Sentinel-2.

Source → target	**Output assigned to target (%)**	**Real patches assigned (%)**	**Edge corr. with source (real pair)**	**Cycle PSNR (dB) / SSIM**
S2 → SPOT	57.3	74.3	0.82 (0.69)	13.7 / 0.47
S2 → SAR	17.7	60.3	0.87 (0.70)	13.7 / 0.47
SPOT → S2	29.3	62.7	0.71 (0.69)	15.4 / 0.51
SPOT → SAR	80.0	60.3	0.82 (0.61)	19.9 / 0.75
SAR → S2	12.0	62.7	0.94 (0.70)	14.2 / 0.48
SAR → SPOT	45.3	74.3	0.92 (0.61)	24.0 / 0.71

- **Content is preserved.** The correlation between the edge maps of the source and of the output is between 0.71 and 0.94, higher than the correlation between the edge maps of the real source and the real co-registered target patches of the other sensor (0.61 to 0.70) in all six pairs. The output thus keeps the structure of the source, which is what the semantic and edge losses aim for.

- **The round trip recovers the source only moderately.** The cycle reconstruction reaches 24.0 dB for SAR to SPOT and back, and 19.9 dB for SPOT to SAR and back, but only 13.7 to 15.4 dB when the source is Sentinel-2 or when the intermediate domain is Sentinel-2.

4.6.4 Reconstruction with the patch's own style

When a patch is conditioned on its own domain, the output should reproduce the patch. Table 4.8 shows that this works for SPOT (20.3 dB) and SAR (24.3 dB) but not for Sentinel-2 (13.7 dB, SSIM 0.48), even though this is the identity mapping that the identity loss trains. Conditioning on the embedding of the patch itself gives similar values as the centroid. Only 57% of the reconstructed Sentinel-2 patches are recognized as Sentinel-2, against about 88% to 90% for SPOT and SAR. The Sentinel-2 domain is thus the weak point of the generator, both as a source and as a target.

TABLE 4.8: Reconstruction of a held-out patch when the generator is conditioned on its own domain (PSNR in dB / SSIM).

Domain	Centroid style	Embedding of the patch itself	Output assigned to own domain (%)
Sentinel-2	13.7 / 0.48	12.5 / 0.52	57.0
SPOT 6/7	20.3 / 0.77	21.4 / 0.76	87.7
SAR	24.3 / 0.74	22.0 / 0.71	89.7

4.6.5 Brightness bias of the generator

To find out why the translations into Sentinel-2 look like a haze and why they agree poorly with the real patches, the mean value of each output channel was compared with that of the real target patches (Table 4.9). For the Sentinel-2 target, the channel means of the output are the same for all three source domains (they agree to within 0.001), and they are far from the real Sentinel-2 means: the output is brighter by 0.37 in the red, 0.25 in the green and 0.59 in the blue channel. The Sentinel-2 style therefore produces an essentially source-independent color cast, in which the content of the source is visible only as faint structure. The output for the SAR-to-SPOT pair has the same kind of error in the blue channel (0.12 against -0.48 for real SPOT patches), which is consistent with the low agreement of that pair in Table 4.6 and with the higher score of the wrong style. For the other pairs the channel means are close to those of the real target patches (largest deviation 0.19, mostly below 0.15), and for SPOT-to-SAR and SAR-to-SAR they are within 0.04.

The cause of the source-independent Sentinel-2 output was not established. Possible contributing factors, none of which was tested, are the low brightness of the Sentinel-2 data, the fact that the centroid embedding of Sentinel-2 is an average that the generator did not see in training, and the short training.

4.6.6 Continuity of the style space

Figure 4.3 shows one SPOT patch translated with a style that moves from one domain centroid to another, $z(\alpha) = (1-\alpha)\bar{z}_a + \alpha\bar{z}_b$ renormalized to unit length. The evaluation used 300 source patches (100 per domain), and its results are summarized in Table 4.10 and Figure 4.4.

TABLE 4.9: Mean value of each channel of the translated patches (300 patches per source domain) and of the real target patches, in $[-1, 1]$. Channels are red, green, blue and near-infrared (or proxy) for the optical domains, and VV, VH, difference and mean for SAR.

Target	**Source**	**Ch. 1**	**Ch. 2**	**Ch. 3**	**Ch. 4**
Sentinel-2	real patches	−0.62	−0.68	−0.77	0.02
	from Sentinel-2	−0.25	−0.43	−0.18	−0.06
	from SPOT 6/7	−0.25	−0.43	−0.18	−0.06
	from SAR	−0.25	−0.43	−0.18	−0.06
SPOT 6/7	real patches	−0.35	−0.23	−0.48	−0.29
	from Sentinel-2	−0.40	−0.34	−0.49	−0.24
	from SPOT 6/7	−0.29	−0.23	−0.34	−0.27
	from SAR	−0.20	−0.23	0.12	−0.26
SAR	real patches	0.17	−0.26	1.00	−0.04
	from Sentinel-2	−0.02	−0.40	0.90	−0.07
	from SPOT 6/7	0.15	−0.27	0.96	−0.06
	from SAR	0.21	−0.25	0.98	−0.05

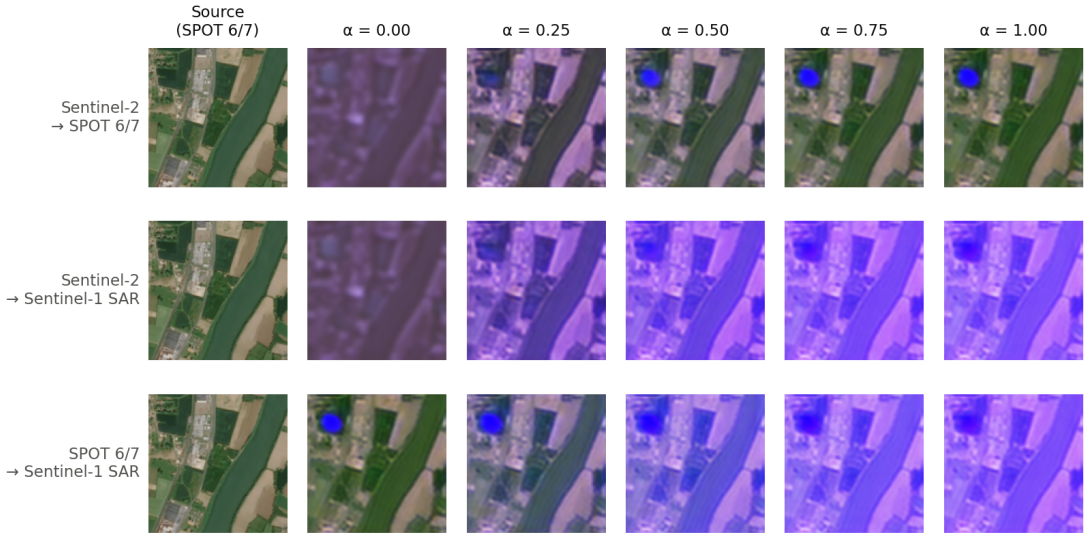


FIGURE 4.3: One SPOT 6/7 patch translated with styles that are interpolated between two domain centroids (rows), for α from 0 to 1.

TABLE 4.10: Response of the output to the interpolation coefficient α (300 source patches). Endpoint L1 is the mean absolute difference between the outputs for $\alpha = 0$ and $\alpha = 1$; the linearity ratio is the endpoint L1 divided by the summed L1 of the four successive steps; the monotone share is the fraction of patches for which the distance to the $\alpha = 0$ output never decreases.

Interpolation	Cosine of centroids	Endpoint L1	Linearity ratio	Monotone share (%)
Sentinel-2 ↔ SPOT 6/7	−0.04	0.122	0.60	41
Sentinel-2 ↔ SAR	0.15	0.238	0.91	96
SPOT 6/7 ↔ SAR	−0.03	0.232	0.96	100

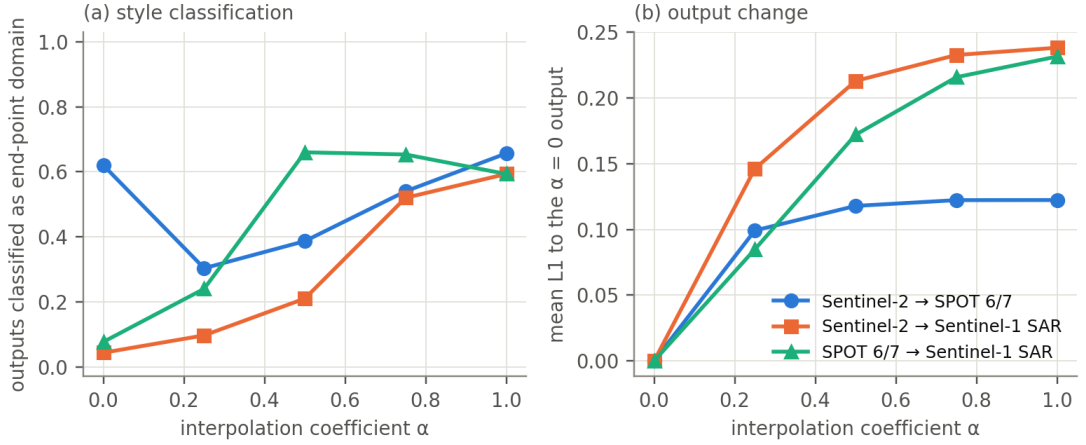


FIGURE 4.4: (a) Fraction of interpolated outputs that the encoder assigns to the end-point domain b; (b) mean absolute difference to the output for $\alpha = 0$.

The output changes smoothly with α when SAR is one of the end points. For the two pairs with SAR, the change is close to linear (linearity ratio 0.91 and 0.96) and monotone for 96% and 100% of the patches, and in the grid the tone moves gradually from green to violet. The style classification in Figure 4.4(a) increases toward the SAR end, although not strictly monotonically. For Sentinel-2 and SPOT the behavior is different: the output changes by only about half as much between the end points (0.122 against 0.23 to 0.24), the change is concentrated in the first step from $\alpha = 0$ to $\alpha = 0.25$ and then saturates (linearity ratio 0.60), and only 41% of the patches change monotonically. The step at $\alpha = 0$ is visible in the top row of Figure 4.3, where the pure Sentinel-2 style yields the haze described above and the first interpolation step already produces a detailed, SPOT-like image. The fractions in Figure 4.4(a) depend on the mixture of the three source domains and should not be read as absolute rates.

4.7 Discussion

4.7.1 Objective 1

The encoder was trained without any domain label and solves its contrastive task on held-out patches from a largely different region. Its embeddings are spread continuously over the hypersphere and can be interpolated, so the objective of a continuous, label-free embedding is met as a design. The embedding is, however, only a moderate descriptor of the sensor: 65.8% of the held-out patches are assigned to the correct sensor, the clusters overlap, and the effective dimensionality is about four. The results suggest, but the experiments did not test, several reasons. The pretext task rewards the discrimination of individual patches, and it has no term that groups the patches of a sensor: the only positive pairs are two augmented views of the same patch, and the negatives include patches of the same sensor. Since the pooled feature statistics are largely invariant to flips and rotations, the encoder can solve the task with patch-specific statistics that depend on the scene as much as on the sensor. The project plan foresaw positive pairs that are defined by the spatial or temporal proximity of patches; this was not implemented. Further factors are the saturated SAR channel, the short schedule of 10 epochs, and the shift between the training regions and the held-out region.

4.7.2 Objective 2

A single generator translates any of the three source domains into any of the three target styles in one forward pass of about 84 ms per patch on the CPU, without domain labels at training time. The results show three properties of the resulting model. It conditions on the style vector: a wrong style gives a lower agreement in five of six pairs, and the output changes smoothly with the interpolation coefficient for the SAR pairs. It preserves structure: the edge correlation with the source is higher than that between real sensor pairs. It does not reproduce the radiometry of the target sensor: a constant image has a higher PSNR and SSIM than the translation, the encoder recognizes the outputs as the target domain only for some pairs, and Sentinel-2 is rendered as a blurred haze. Translation into SAR from SPOT and into SPOT from Sentinel-2 is the most successful, and all translations that involve Sentinel-2 are the least successful.

The Sentinel-2 problem is probably related to the data and the training. The Sentinel-2 patches are the darkest and have the lowest contrast in the data (Table 4.2), and the identity mapping of Sentinel-2 is also poorly reproduced (13.7 dB), which indicates that the generator has not learned this domain well within its 12 epochs and its narrow network. The generator is also trained with instance embeddings of random target exemplars but evaluated here with centroids; centroids are an average of many patches and are not necessarily in the region of the embedding space that the generator has seen, which may explain why they behave differently from instance embeddings for some pairs.

4.7.3 Relation to the related work

Table 2.2 placed the target of this work in relation to the literature on four properties. In the implementation, the training uses no domain labels, one generator serves all ordered pairs of three domains, the domain code is a continuous 64-dimensional embedding that can be interpolated, and the data are satellite imagery. In this sense the design goal of Objectives 1 and 2 is met. The quality of the result, however, cannot be compared with the cited methods: none of them was trained on the same data, and the published figures were obtained on other datasets and tasks. It is therefore not claimed that the proposed approach is better than SS(DA)^2, StarGAN v2 or the other methods.

4.7.4 Summary of the achievement of the objectives

Table 4.11 summarizes the extent to which each objective is met.

4.8 Limitations and threats to validity

The results above should be read with the following limitations in mind.

L1. Region shift in the held-out set. All 300 held-out patches come from one Sentinel-2 tile (T31TFM), while the training set is dominated by two other tiles (Section 3.3). The held-out results therefore combine the effect of the models with the effect of a region shift. In particular, the domain centroids

TABLE 4.11: Achievement of the objectives.

Objective	Achieved	Not (yet) achieved
1. Domain encoder	Designed, implemented and trained without labels; continuous, non-collapsed embedding; solves the pretext task on held-out regions	Sensor identity is captured only moderately (65.8% against 33.3% chance); no cross-sensor content signal; low effective dimensionality
2. Conditional generator	One model for all ordered domain pairs; conditioned on a continuous embedding; structure preserved; smooth response for SAR pairs; fast inference	Target radiometry not reproduced; translations into Sentinel-2 fail; pixel agreement below a constant baseline

and the reference embeddings for the nearest-neighbor classifier come from mostly different regions than the patches that they are used on.

L2. Modalities that are not simultaneous. The paired references differ in acquisition date from the source patches, by about 11 days for Sentinel-1 and by months for SPOT. Pixel-level agreement with the reference (PSNR, SSIM, L1) is therefore bounded, even for a perfect translator, and it is informative mainly in comparison with the reference baselines.

L3. Preprocessing artifacts. The saturated SAR difference channel and the synthetic SPOT fourth channel (Section 4.4) reduce the information that the models can use, and they may influence how well some domains can be told apart.

L4. Limits of the metrics. PSNR and SSIM measure agreement with a co-registered image and not realism. The style-transfer success rate is judged by the same encoder that conditions the generator, so it can only be as reliable as that encoder, which Section 4.5 shows to be a moderate domain classifier. Perceptual metrics that need networks pretrained on natural images, and a human evaluation, were not used.

L5. One training run. Each model was trained once, without a fixed seed, so the variation between training runs is unknown; the reported standard deviations describe variation across patches only. No ablation of the six generator loss terms was run, so the effect of an individual loss cannot be attributed.

L6. Small compute budget and unused data. The networks are narrow and were trained for 10 and 12 epochs on a CPU. The unlabeled MSC-France corpus that the plan foresees for the encoder was not used.

L7. Use of domain labels at evaluation. Domain centroids, the classification accuracies and the silhouette coefficient use the domain of each patch. They were used only to evaluate the models, and never for training, but a deployment without any domain information would have to obtain the target style from an exemplar patch instead of a centroid.

L8. No baseline. No other translation method (for example a pairwise CycleGAN-type model, SS(DA)^2 or StarGAN v2) was trained on the same data, so no statement about relative performance can be made.

Chapter 5

CONCLUSION

This chapter summarizes the work, states the conclusions in relation to the two objectives, lists the contributions, and describes the scope for future research and development.

5.1 Summary of the work

The project addresses the degradation of deep learning models for satellite imagery when they are applied to another sensor than the one that they were trained on. The literature review (Chapter 2) showed that the existing translation-based remedies either need domain labels, are limited to a pair of domains, or handle one target at a time, and that self-supervised pretraining for remote sensing has not been applied to represent the difference between sensors. This report addressed the first two of the four project objectives. A two-stage framework was designed and implemented (Chapter 3): a domain encoder that is trained with a contrastive objective on unlabeled patches and produces a continuous 64-dimensional embedding, and a single AdaIN-conditioned generator with a projection discriminator that translates a source patch into the style of a target embedding under a six-term objective. The framework was trained on a subset of the PASTIS-HD benchmark with three sensors (Sentinel-2, SPOT 6/7 and Sentinel-1 SAR), 1,500 patches per sensor, on a CPU, and it was evaluated on 300 held-out patches per sensor from a region that differs from most of the training regions (Chapter 4).

5.2 Conclusions with respect to the objectives

5.2.1 Objective 1: domain encoder

To design and develop a domain encoder that learns continuous-valued domain embeddings directly from unlabeled multi-sensor satellite image patches.

A domain encoder with 888,224 parameters was designed, implemented and trained without any domain label. It solves its contrastive task on held-out patches from a largely different region (a held-out loss of 0.24 against 2.08 for an uninformative embedding), and its embeddings are spread continuously over the hypersphere instead of collapsing into a few clusters, so that they can be interpolated. The embedding, however, describes the sensor only moderately: 65.8% of the held-out patches are assigned to the correct sensor by the nearest centroid, against 33.3% by chance, the sensors overlap in the embedding space (silhouette coefficient 0.11), and the effective dimensionality is about four. It can be concluded that contrastive learning on augmented views of the same patch is sufficient to obtain a label-free and continuous embedding, but not sufficient to obtain a strong sensor descriptor. The objective is therefore met as a design and an implementation, and only partly met with respect to the quality of the sensor information in the embedding.

5.2.2 Objective 2: conditional generator

To develop a conditional generator that translates image content into the style of an arbitrary target domain using a single unified model.

A single generator with 1,594,116 parameters was developed that translates any of the three sensor domains into any other in one forward pass of about 84 ms per patch on the CPU. The evaluation showed that it uses the style vector (a wrong style is worse for five of six pairs, and the output changes smoothly with an interpolated style for the pairs with SAR), and that it preserves the structure of the source (edge correlation of 0.71 to 0.94 against 0.61 to 0.70 between real sensor pairs). It does not reproduce the radiometry of the target sensor: a constant image with the mean of the target domain agrees better with the real co-registered patch than the translation, the encoder recognizes the outputs as the target domain in

12% to 80% of the cases depending on the pair, and every translation into Sentinel-2 gives a source-independent color cast instead of a Sentinel-2-like patch. The objective is therefore met for the architecture, since one model serves all ordered domain pairs and takes an arbitrary embedding as target, and for translations into SAR and SPOT; it is not met for translations into Sentinel-2, and the quality of the translation is limited overall.

5.3 Contributions

- An implementation of a label-free, single-model framework for multi-sensor satellite image translation that consists of a contrastive domain encoder, an AdaIN-conditioned generator, a style-projection discriminator and a six-term objective that constrains content.
- A reproducible held-out evaluation protocol for such a framework, which uses co-registered references, trivial baselines and a wrong-style control, and which measures both the embedding and the translation, together with the scripts that implement it.
- An analysis of the properties of the PASTIS-HD data that affect such experiments: a split that follows the identifier order and therefore separates the regions, modalities that are not simultaneous, and a saturated SAR channel.
- Evidence about the limits of the first version: the encoder captures the sensor only moderately, and the generator does not reproduce the target radiometry, in particular for Sentinel-2.

5.4 Scope for future research and development

The results define the following directions.

1. **Remaining project objectives.** Evaluate the generalization of the framework to sensor pairs that were never observed together in training, with a leave-one-pair-out protocol on MultiEarth 2022 and a zero-shot test on SpaceNet 6, and evaluate the effect of the translation on downstream classification and segmentation accuracy on PASTIS-HD and MultiEarth. These

evaluations should include a pairwise baseline such as CycleGAN or SS(DA)^2, and StarGAN v2, trained on the same data.

2. **A pretext task that captures the sensor.** Replace or complement the augmented-view positives with the spatial and temporal proximity positives that the project plan foresaw, and add sensor-aware negatives, so that patches of the same sensor are grouped in the embedding space. Pretrain on the unlabeled MSC-France corpus, which adds Landsat-8 and a much larger number of locations.

3. **Data and preprocessing.** Use all 2,433 PASTIS-HD patches with a split that is stratified by tile, select the Sentinel-1 acquisition that is closest in time to the Sentinel-2 acquisition, rescale the SAR difference channel so that it does not saturate, replace the synthetic SPOT fourth channel by a sensor-specific channel adapter, and rescale Sentinel-2 so that its contrast is comparable to the other sensors.

4. **Training of the generator.** Train with wider networks and longer schedules on a GPU, train with centroid-like styles in addition to instance embeddings so that the styles used at inference are also seen in training, run ablations of the six loss terms, and repeat the training with different seeds to measure the variation between runs. Seed and log the training runs so that they can be reproduced.

5. **Style representation.** Reduce the patch-specific content in the embedding, for example with an adversarial or a regularizing term, and study whether a lower-dimensional or a multi-prototype description of each sensor is better suited than a single centroid.

6. **Extensions named in the project reviews.** Cross-modality adaptation between optical and SAR imagery, combination with pretrained remote sensing foundation models such as SatMAE or Prithvi, application to bitemporal change detection, and an evaluation of the adversarial robustness of the adaptation pipeline. The viewing-angle domains that are already implemented in the framework can be evaluated as well.

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Appendix A

Layer-wise Network Specification

The tables below list the layers of the three networks as implemented. "Conv $k \times k$, s" denotes a convolution with kernel size k and stride s; IN is instance normalization; LReLU is a LeakyReLU with slope 0.2; SN is spectral normalization. Sizes are given as channels × height × width for an input patch of $4 \times 128 \times 128$.

TABLE A.1: Domain encoder E (888,224 trainable parameters).

Block	**Operation**	**Output size**
Input adapter	Conv 3×3, s1, 4→32; LReLU	$32 \times 128 \times 128$
conv1	Conv 4×4, s2, 32→64; IN; LReLU	$64 \times 64 \times 64$
conv2	Conv 4×4, s2, 64→128; IN; LReLU	$128 \times 32 \times 32$
conv3	Conv 4×4, s2, 128→256; IN; LReLU	$256 \times 16 \times 16$
SE block	Global average pooling; FC 256→32; ReLU; FC 32→256; sigmoid; channel-wise scaling	$256 \times 16 \times 16$
Statistics	Channel-wise mean and standard deviation, concatenated	512
Angle branch	FC 1→32; LReLU; FC 32→32 (zero vector if no angle is given)	32
Projection head	FC 544→256; LReLU; dropout 0.1; FC 256→128; LReLU; FC 128→64	64
Normalization	L2 normalization	64

TABLE A.2: Generator G (1,594,116 trainable parameters). Each AdaIN layer has two linear maps, $64 \to C$, for γ and β.

Block	**Operation**	**Output size**
enc_conv1	Reflection pad 3; Conv 7×7, s1, 4→32; IN; LReLU	$32 \times 128 \times 128$
enc_conv2	Conv 4×4, s2, 32→64; IN; LReLU	$64 \times 64 \times 64$
enc_conv3	Conv 4×4, s2, 64→128; IN; LReLU (content code c)	$128 \times 32 \times 32$
Residual block $\times 4$	Conv 3×3 (no bias); AdaIN; LReLU; Conv 3×3 (no bias); AdaIN; skip connection	$128 \times 32 \times 32$
Decoder 1	Bilinear upsampling $\times 2$; Conv 3×3, 128→64; AdaIN; LReLU	$64 \times 64 \times 64$
Decoder 2	Bilinear upsampling $\times 2$; Conv 3×3, 64→32; AdaIN; LReLU	$32 \times 128 \times 128$
Output	Reflection pad 3; Conv 7×7, 32→4; tanh	$4 \times 128 \times 128$

TABLE A.3: Discriminator D with style projection (709,601 trainable parameters).

Block	**Operation**	**Output size**
conv1	SN Conv 4×4, s2, 4→32; LReLU	$32 \times 64 \times 64$
conv2	SN Conv 4×4, s2, 32→64; IN; LReLU	$64 \times 32 \times 32$
conv3	SN Conv 4×4, s2, 64→128; IN; LReLU	$128 \times 16 \times 16$
conv4	SN Conv 4×4, s1, 128→256; IN; LReLU	$256 \times 15 \times 15$
Unconditional head	SN Conv 3×3, 256→1	$1 \times 15 \times 15$
Style projection	FC 64→256; inner product with the features, summed over channels and divided by $\sqrt{256}$; added to the unconditional output	$1 \times 15 \times 15$

Appendix B

Code Organization and Reproducibility

B.1 Organization of the code base

The implementation is organized as follows.

- `configs/config.py`: default configuration and the dataset-specific settings for PASTIS-HD.
- `data/`: the converter from PASTIS-HD to the unified format (`convert_real_datasets.py`), the dataset class (`dataset.py`), the augmentations (`transforms.py`), the dataset downloader, and the synthetic-scene generator used for development.
- `models/`: the domain encoder and the contrastive loss (`domain_encoder.py`), the generator (`generator.py`), the discriminator (`discriminator.py`) and the generator losses (`losses.py`).
- `train_domain_encoder.py` and `train_generator.py`: Stage 1 and Stage 2 of the training.
- `tests/`: 29 automated tests.
- `report/scripts/eval_objectives_1_2.py`: the held-out evaluation of Chapter 4; it writes all metrics to `report/results/metrics.json`.

- `report/scripts/make_figures.py`: regenerates the figures of Chapter 4 with the same checkpoints and the same protocol.
- `report/scripts/eval_brightness_bias.py`: the per-channel mean comparison of Table 4.9; it writes `report/results/brightness_bias.json`.

B.2 Commands

The commands below are run from the root directory of the project. The trained checkpoints evaluated in this report are stored in `outputs_pastis_fixed/`; a configuration file that sets `output_dir` to this directory redirects the checkpoints of the training stages there.

```
# convert the first 1500 aligned PASTIS-HD patches into the unified format
python data/convert_real_datasets.py --pastis_dir data/PASTIS-HD \
       --output_dir data/france_dataset --max_patches 1500

# Stage 1 (Objective 1) and Stage 2 (Objective 2)
python main.py --dataset pastis --stage encoder   --config my_config.yaml
python main.py --dataset pastis --stage generator --config my_config.yaml

# held-out evaluation used in Chapter 4 (about 30 minutes on the CPU described ↩
    in Table 3.4)
python report/scripts/eval_objectives_1_2.py
python report/scripts/eval_brightness_bias.py
python report/scripts/make_figures.py

# unit tests
python -m pytest -q
```

where `my_config.yaml` contains the single line `output_dir: outputs_pastis_fixed`.

B.3 What can and cannot be verified from the stored artifacts

The tensor shapes stored in the checkpoints confirm the architecture of Table 3.3 and Appendix A (base channel width 32, embedding dimension 64, four residual blocks). The training logs of the run that produced the checkpoints were not saved, so the number of epochs and the batch size in Table 3.3 are the values that

the pipeline defines for the PASTIS-HD dataset, and they could not be checked against the run itself. Per-epoch loss curves are therefore not reported. As noted in Section 3.5, the training loops are not seeded, whereas the evaluation script is.